

A diagnostic model for hydraulic fracture in naturally fractured reservoir utilising water-hammer signal

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ARTICLE INFO

Keywords:

Hydraulic fracture
Water-hammer in wellbore
Fracture diagnostic
Natural fracture

ABSTRACT

The diagnostic of hydraulic fractures is vital to the exploitation of subsurface resource. Diagnostic technique for hydraulic fracture based on the water-hammer pressure have been gradually highlighted owing to their cost effectiveness and simplicity. The present diagnostic models overlook the effects of fluid leak-off and natural fracture in hydraulic fractures, and it is limited for application in naturally fractured reservoirs. In this study, the location and number of hydraulic fractures are first obtained through the enhancement and cepstrum processing of a water-hammer signal. Subsequently, the water-hammer pressure within the wellbore is calculated by solving the continuity and momentum equations for the fluid. Wellbore and hydraulic fractures are considered as a hydraulic system. To estimate the fracture dimension, flow boundary conditions are imposed to the fluid leak-off, interactions between natural and hydraulic fractures, and multifracture stress shadows. The results show that the fracturing shut-in method can be appropriately adjusted to avoid large pressure pulsations, which damage well integrity, and to obtain a clear water-hammer signal for fracture diagnosis. Natural fractures reduce the hydraulic fracture dimensions but facilitate the creation of complex fracture networks, while this complexity cannot be increased indefinitely. The minimum horizontal stress decreases the fracture dimension and a greater difference in the horizontal stress renders it easier for hydraulic fractures to cross natural fractures to create larger dimensions. The field study shows the optimisation measures can be recommended based on the diagnostic results.

1. Introduction

In hydraulic fracturing, hydraulic fractures are created by injecting fluids into a wellbore to break a rock structure. Hydraulic fractures are vital to the exploitation of subsurface resource (oil, natural gas, geothermal, etc) using hydraulic fractures to increase the permeability of reservoirs [1,2]. Hydraulic fracture dimensions must be assessed via fracturing-diagnostic techniques, which are essential for evaluating the treatment performance, optimising the treatment method, and guiding reservoir engineering design. The

Abbreviations: AHFM, algebraic heat flux model; ISIP, Instantaneous shut-in pressure; Re, The Reynolds number; RCI, The diagnostic model based on R, C and I; MOC, Method of characteristics; EMD, Empirical mode decomposition; IMFs, Intrinsic mode functions; IMF, The decomposition component of IMFs; CEEMD, Complementary ensemble empirical-mode decomposition; E-frac, Equivalent hydraulic fracture; Δx , Spatial mesh; Δt , Temporal mesh.

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<https://doi.org/10.1016/j.engfracmech.2025.111347>

Received 9 October 2024; Received in revised form 11 June 2025; Accepted 14 June 2025

Available online 16 June 2025

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Nomenclature

a	Speed of the water-hammer wave
L	Well depth
t	Time after shut-in
u_0	Steady flow rate
$P(z)$	Pressure distribution in wellbore
$x(t)$	Surface pressure signal
$e(t)$	The noise components in pressure signal
$x_0(t)$	The useful pressure signals
k	The number of IMFs
$r_k(t)$	Final residual value
R_n	Correlation coefficients of decomposition component
$\hat{x}_0(\tau)$	The cepstrum of the water-hammer source signal
τ	Frequency of signal
ω	The angular frequency
V_i	The cepstrum value with different quefrency times
V_j	The cepstrum value with different physical times
F_{ij}	The cepstrum eigenvalues
K	Bulk modulus of the fluid
ρ	Fluid density
E	The pipe Young's modulus of elasticity
D	Pipe diameter
E	The pipe wall thickness
A_{pipe}	The cross-sectional area of the fracturing pipe string
u	Velocity of fluid
E_{pipe}	Young's modulus of pipe
H	Pressure head
θ	The angle between the pipe axis and gravitational acceleration
τ_{xy}	Deviatoric stress
λ	Friction factor
δ	The wall roughness of the fracturing pipe string
$\bar{u}$	The average velocity of fluid
Q	Flow rate
P_{bh}	Bottomhole pressure
$\bar{P}_f$	The average near-wellbore pressure
P_{pore}	Pore pressure of the reservoir
ΔP_{nf}	The additional natural fracture pressure drop
P_{net}	The net pressure of the hydraulic fracture
$\overline{P_{net}}$	The average net pressure
ΔP_{shadow}	The stress shadow in multifracture
P_{end}	The pressure at the end of the water hammer
γ	Damping factor of the water hammer
M	Magnitude of the water hammer
ω	Frequency of the water hammer
α	Phase of the water hammer
b	Magnitude of the steady exponential pressure decay
m	Decay factor of water hammer
c	Steady-state value of water hammer
N_{nf}	Number of natural fracture
ρ_{nf}	Density of natural fracture
A_{re}	Area of the reservoir
W_{nf}	The natural fracture width
L_{nf}	The natural fracture length
W_{aver}	The average width of natural fracture
L_{aver}	The average length of natural fracture
σ_W^2	Standard deviation of the width of natural fracture
σ_L^2	Standard deviation of the length of natural fracture

V_{leak}	The leak-off total volume
C_{leak}	Leak-off coefficient
W_f	Width of hydraulic fracture
H_f	Height of hydraulic fracture
L_f	Half-length of hydraulic fracture
θ_{nf}	Strike angle of natural fracture
A_{nf}	The cross-sectional area of natural fracture
θ_{nf}	Strike angle of the natural fracture
θ_{aver}	The average strike angle of natural fracture
K_θ	Deviation coefficient
K_{rand}	Random number
Q_0	The steady-state flow rate prior to fracturing shut-in
R	Resistance of fracture system
C	Capacitance of fracture system
C_n	Capacitance of the n -th hydraulic fracture
I	Inertance of fracture system
S_h	Horizontal minimum stress
E_{rock}	Young's modulus of rock
ν	Poisson's ratio of rock
$E(m)$	The complete elliptical integral of the second type
$W_{f,n}$	The width of the n -th propagated hydraulic fracture
$L_{f,n}$	The length of the n -th propagated hydraulic fracture

most typically utilised hydraulic fracture-diagnostic techniques in reservoirs include microseismic modelling, distributed fibre-optic sensing, wide-field electromagnetic methods, tracers, wellbore imaging, and well testing [3–6]. However, these methods may be cost prohibitive and may not necessarily achieve non-intrusive real-time monitoring. A water-hammer signal is a period of pressure pulsation obtained on a surface after fracturing shut-in and contains information regarding the wellbore and hydraulic fracture. Therefore, diagnostic methods based on the wellhead water-hammer pressure are gradually highlighted owing to their cost effectiveness and simplicity [7–9].

Holzhausen [10] initially introduced the basic principles for analysing the dimensions of hydraulic fractures through surface-pressure pulsation. The dimensions of a hydraulic fracture and its connectivity to the bottomhole affect the hydraulic pulsation properties within the wellbore. Measuring this effect is an uncomplicated and cost-effective diagnostic method for identifying the associated hydraulic fracture. Holzhausen [10] developed a hydraulic representation of a fracture crossing a bottomhole by incorporating two combined factors: flow resistance and capacitive fracture. Field measurements of hydraulic impedance testing in an injection well further validated the feasibility of using the water-hammer pulsation pressure for fracturing diagnostics [11,12]. Subsequently, researchers have developed and applied this method. Yew [13] expanded the method by applying a Fourier inverse transform to wellhead data to estimate the fracture dimension. Patzek [14] proposed a model for lossy transmission lines that incorporated the energy depletion of pressure waves to characterise the dimensions of a wellbore and fractures. Furthermore, Carey [7,15] suggested a model for resistance–capacitance–inertance (*RCI*) fractures, which was characterised based on the resistance, capacitance, and inertance via the analogy of hydraulic and electrical systems. The *RCI* model was used as a boundary condition in wellbore water-hammer pressure calculation, thus enabling the diagnosis of fractures with the same dimensions. However, the fracture lengths estimated under the water-hammer *RCI* mode exceeded those calculated using alternative techniques (microseismic, electromagnetic imaging, pressure interference testing, and long-term production data analysis), although they were comparable to those derived based on microseismic stimulated reservoirs volumes and the fracture complexity index [16,17]. This is because the mechanisms of water-hammer signals and the associated characteristic parameters (R , C , and I) are complex. Furthermore, they are related to factors such as the types of wells completed, the wellbore dimensions, the fluid properties, and the physical properties of the rocks surrounding the fracture [8,18,19].

The *RCI* model is based on the ideal assumptions that the fractures are uniformly and efficiently propagated, the fracturing fluid leak-off during water hammer is zero, and the hydraulic fracture is a symmetric biplane. The leak-off of fracturing fluid is the main factor that determines the hydraulic fracture dimension [20]. Therefore, additional enhancements are required. In study of [18], a two-parameter fracture boundary condition model considering fracturing fluid leak-off was proposed to estimate the fracture dimensions. The leak-off characteristics were determined by fitting the end curve of the water hammer, and the net pressure drop at the bottomhole was defined as a function of the leak-off volume. Hwang [21] reported that stress shadowing between multifracture changed the stress distribution, thus affecting the fracture-dimension diagnostics from water-hammer analysis. Therefore, Hu [22] proposed a multifracture diagnostic model by first considering the stress shadow between hydraulic fractures using the *RCI* fracture model, and then determined whether the hydraulic fractures were effectively propagated by assessing the positive and negative average net pressures in multifracture. However, this process does not consider the effect of fracturing fluid leak-off and natural fractures on the fracture dimensions, which is limited for application in naturally fractured reservoirs.

Another application of water-hammer pressure pulsation in fracturing diagnostics involves assessing the status of fluid injection in

each perforation cluster [9,23,24]. A series of signal analysis methods was employed on high frequency water-hammer signals to derive the eigenvalues output by perforation clusters in absorbing fluids. The position based on the eigenvalue was identified as the perforation cluster that opened effectively owing to the absorbing fracturing fluid. This provides ideas for us to determine the location and number of hydraulic fractures in the bottomhole using our model. Based on field experience, this method is applicable to large-cluster space fracturing [9]. Nonetheless, identifying the precise locations of perforation clusters remains challenging, particularly for high frequency signals, owing to noise effects. A precise model for signal analysis is essential for driving the target information in the field water-hammer pressure signals. Korkin [25] developed a simple algorithm and a cepstrum matrix to select a suitable calculation region in recorded data and to enhance the fracture depth information in high frequency water-hammer pressure signals. Li [26] proposed a comprehensive filtering model for water-hammer pressure signals using different filtering methods, including the Hampel filter, finite impulse-response low-pass filter, adaptive notch, and median filter, which are effective for water-hammer signals from laboratory simulations. A suitable signal-enhancement method is required for more complex field water-hammer signals.

Resources in unconventional reservoirs such as shale oil and gas have become important sources of fossil energy [27,28]. Unconventional reservoirs are mostly naturally fractured reservoirs. Ultralow permeability and the wide distribution of natural fractures are the two main characteristics of unconventional reservoirs [29–31]. Therefore, multicluster fracturing of horizontal wells has become a key technique for increasing stimulated reservoir volume [32–34]. However, studies showed that not every perforation cluster is opened to allow the propagation of hydraulic fractures owing to the inhomogeneous perforation quality, reservoirs properties, and stress [35,36]. Meanwhile, hydraulic fractures were shown to propagate with different dimensions because of the stress shadow of multiple clusters and the inhomogeneity of reservoirs mechanical and in-situ stresses [37,38]. In addition, natural fractures significantly affected the propagation and hydraulic fracture dimensions. Hydraulic fractures can open or penetrate natural fractures, thus creating complex fracture networks and greater pressure loss, and the fracturing fluid can enter natural fractures, thereby significantly increasing the leak-off volume [2,32,39]. In water-hammer studies, one must determine the location and number of hydraulic fractures resulting from the opening of perforation clusters, in addition to the effect of natural fractures. In conclusion, although the fracturing diagnostic technique based on water-hammer pressure analysis has been successfully applied, its reliability in identifying fractures in naturally fractured reservoirs is questionable, owing to some limitations of the current model. Therefore, developing a diagnostic model for multiple hydraulic fractures in naturally fractured reservoir is essential.

In the abovementioned diagnostic models, the effects of fluid leak-off and natural fractures were overlooked, thus rendering it challenging to map hydraulic fractures in naturally fractured reservoir. In this study, a diagnostic model considering the location, number, and dimensions of hydraulic fractures is developed. The location and number of hydraulic fractures are first obtained through the enhancement and cepstrum processing of a wellhead high frequency water-hammer signal. Subsequently, the wellbore water-hammer pressure is calculated by solving fluid-continuity and momentum equations. An RCI multifracture dimension estimation model is developed by coupling the fracturing fluid leak-off, natural and hydraulic fracture interactions, and multifracture stress shadows. Water-hammer pressures with different shut-in modes and multifracture dimensions with different perforation designs, natural fracture densities, and in-situ horizontal stresses are investigated. Finally, a case study of a field shale well is performed to demonstrate the application process and performance of our model.

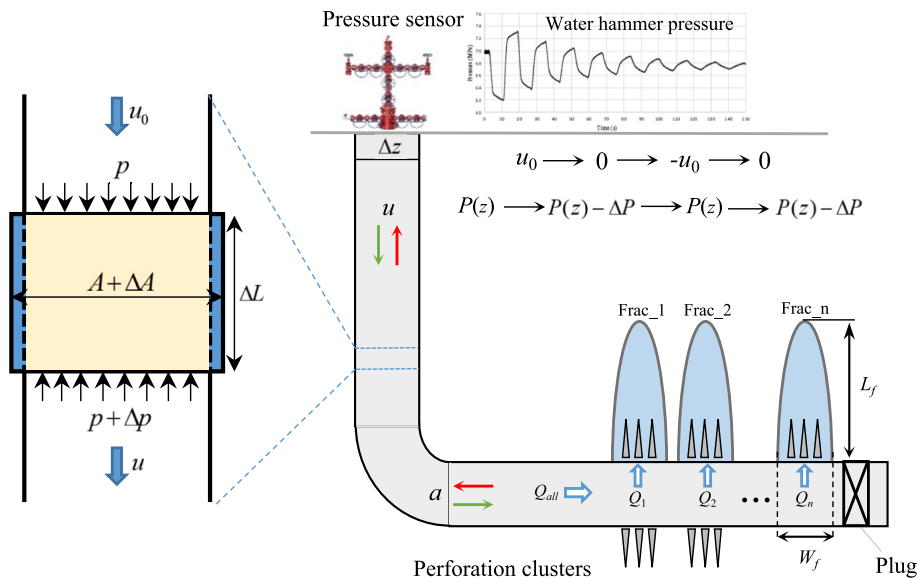


Fig. 1. Schematic diagram of fluid flow and water-hammer signal after shut-in of fracturing treatment.

2. Methodology

2.1. Physical model and water-hammer pressure signal

Fracturing fluid was injected into a deep reservoir via a pipe string as the working fluid during the fracturing treatment process to create multiple hydraulic fractures for enhancing the fluid permeability. Following the shut-in of the fracturing treatment, a water hammer was formed, and the pressure pulsation signal was subsequently measured at the surface. The generation and characteristics of the water-hammer signals must be clarified for fracture diagnosis. Water hammer occurs when the flow rate changes abruptly owing to valve opening or closing in the pipeline, thus resulting in a transient change in the fluid pressure system. The pressure wave dissipates through the pipe and decays continuously under the effects of viscous forces, flow resistance, and other forces. Fig. 1 presents a schematic illustration of the fracturing fluid flow and water-hammer signal. The symbols a , L , t , u_0 , and $P(z)$ denote the speed of the water-hammer wave (m/s), well depth (m), time of shut-in (s), steady flow rate (m/s), and pressure distribution (Pa) before shut-in, respectively.

The generation of an ideal water-hammer pressure signal can be described as follows: (1) At $0 \leq t < L/a$, the fluid in the wellhead cell Δz features a zero velocity, whereas the fluid in the downstream cell still flows with velocity u_0 when the valve is closed. This results in an instantaneous decrease ΔP in fluid pressure in the upstream cell, thus initiating the propagation of the water-hammer wave from the wellhead to the bottomhole. (2) At $L/a \leq t < 2L/a$, when the water-hammer wave arrives at the bottomhole, the fluid velocity within the, bottomhole cell becomes zero, and the bottomhole pressure becomes less than the reservoir pressure. Subsequently, the fluid flows into the wellbore through the hydraulic fracture, thus causing a change in the flow rate of the bottomhole cell from zero to $-u_0$. Consequently, the bottomhole cell wellbore experiences an increase in pressure ΔP , and the water-hammer wave begins to propagate upward from the bottomhole toward the wellhead. (3) At $t = 2L/a$, the flow rate in the wellhead cell changes from its initial value to zero because the valve is closed when the water-hammer wave arrives at the wellhead, and the process of step (1) is repeated. Eventually, the water-hammer pressure wave gradually dissipates owing to flow friction.

2.2. Multifracture diagnostic model

A diagnostic model for multifracture and its solutions are shown in Fig. 2. First, the number and location of the hydraulic fractures were obtained via the enhancement and cepstrum analysis of the surface high frequency water-hammer signal. Subsequently, the water hammer was simulated using a multifracture RCI model that considered fluid leak-off in the fractures, natural fractures, and stress shadows as boundary conditions. The pressure resulting from the water hammer in the wellbore was simulated numerically through the solution of fluid-continuity and momentum equations using the method of characteristics (MOC). Finally, RCI values obtained iteratively were used to estimate the multifracture dimension. The model adheres to the following basic assumptions:

- (1) The flow is isothermal and disregards the heat transfer among the fluid, formation, and wellbore, which implies that the physical properties of the fluid (density and viscosity) are constant.
- (2) A single perforation cluster initiates a solitary hydraulic fracture, thus suggesting that the number and location of perforation clusters with the absorbing fluid correspond to the number and location of the propagated hydraulic fractures, respectively.
- (3) Each effectively propagated hydraulic fracture is a constant-height planar fracture with an elliptical cross-section. The distribution of natural fractures is consistent across different two-dimensional spaces.

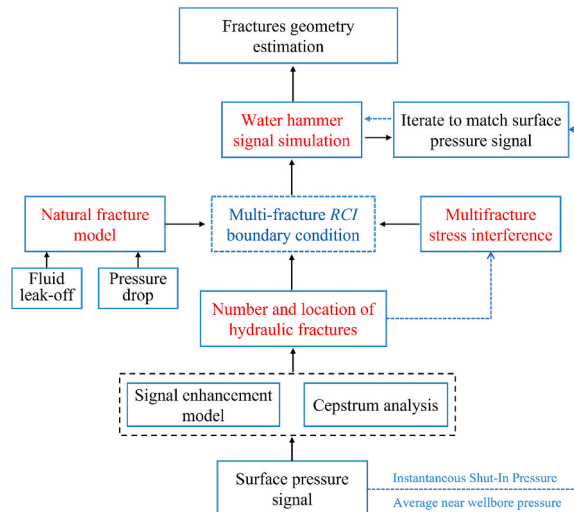


Fig. 2. Structure and solution of multifracture diagnostic model.

- (4) Alterations in the fracture width are the sole cause of dimensional changes occurring during water hammer.

2.2.1. Number and location of hydraulic fracture analysis model

The high frequency pressure signals (≥ 200 Hz) gathered from the surface require denoising followed by cepstrum analysis to identify fracture information, as depicted in Fig. 3.

Signal enhancement model. The surface pressure signal $x(t)$ is composed of intricate noise components $e(t)$. To enhance the useful signal $x_0(t)$, one must minimise the noise effect, which is expressed as Eq. (1). The noise components entail broadband electronic noise, narrow harmonic peaks caused by hydraulic pump strokes, and a zero-frequency trend component linked to decelerate treatment pressure fluctuations [9]. The useful signal is limited to a low-frequency spectrum of 0.1–20 Hz.

$$x_0(t) = x(t) - e(t) \quad (1)$$

The predominant approaches for signal enhancement in signal processing are the short-time Fourier transform, window function filter, wavelet transform, and empirical mode decomposition (EMD). EMD is particularly commendable because it is unconstrained by window functions or wave bases. This renders it adaptable to decompositions, thus resulting in more representative characteristics of the signal investigated [40]. EMD separates a signal into instantaneous frequency components known as intrinsic mode functions (IMFs). White noise pairs are included to ensure continuity and overcome modal mixing resulting from signal singularities during EMD. This approach is known as complementary ensemble empirical-mode decomposition (CEEMD). A detailed description of the CEEMD process is presented in Appendix A. The initial IMF captures the higher-frequency components, whereas the subsequent IMFs capture the lower average frequencies. Summing the IMF components and residual values recovers the original water-hammer pressure signal as follows:

$$x(t) = \sum_m^k IMF_m(t) + r_k(t) \quad (2)$$

where, k is the IMF number and $r_k(t)$ is the final residual value.

Therefore, IMFs may be selectively reconstructed to eliminate unnecessary interference and yield an improved pressure signal related to water hammer. Once the high frequency signal has been removed, the selection of IMFs depends on their correlation coefficients R_n with respect to the original signal, as expressed in Eq. (3). IMFs with $R_n > 0.5$ are typically retained and considered to be well correlated with the original signal.

$$R_n = \frac{\int imf_n(t) \cdot s(t-f)dt}{[\int imf_n^2(t)dt \cdot \int s_n^2(t)dt]} \quad (3)$$

Cepstrum analysis. The useful signal $x_0(t)$ can be represented as the convolution of the source pressure pulse excited by the water hammer and the wellbore reflectivity [9]. Pressure waves reflect at diverse hydraulic fractures, and following the shut-in of multi-cluster fracturing, different signals are blended. It Notably, the reflectivity of hydraulic fracture is negative. Thus, information regarding hydraulic fractures must be acquired using signal-processing methods to determine the reflectivity in the convolved signal. Cepstrum analysis is a widely adopted technique to process convolved signals for identifying signals associated with pipeline leaks [41]. It is mathematically expressed as follows:

$$\hat{x}_0(\tau) = \frac{1}{2\pi} \int \log \left[\left| \int x_0(t) e^{i\omega t} dt \right| \right] e^{-i\omega \tau} d\omega \quad (4)$$

here, $\hat{x}_0(\tau)$ is the cepstrum of the water-hammer source signal and the reflectivity of the wellbore, τ is the frequency, which has the dimension of time, and ω is the angular frequency. During this process, the Fourier transform converts the convolutional signal into a multiplicative signal before performing a logarithmic operation to generate a summation signal. Finally, the inverse Fourier transform is applied to return to the time domain. A cepstrogram of the water-hammer pressure was calculated by integrating the cepstrum results with a series of windows. The pressure oscillations resulting from tube-wave reflections caused by hydraulic fractures appeared as strong negative peaks in the cepstrogram at certain frequencies. Hence, the deepest minima of the cepstrogram in this area can be used to identify hydraulic fractures. However, accurately identifying hydraulic fractures in this region may be challenging. A spectrogram is typically used to extract the negative cepstrum matrix. In this study, we calculated the cepstrum eigenvalues at different frequencies based on the methodology developed by [25] and [42] as follows:

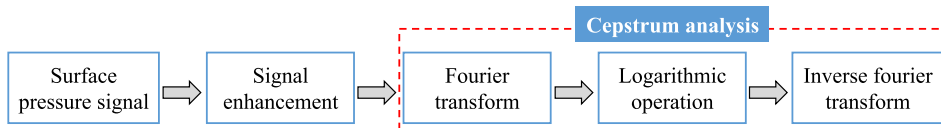


Fig. 3. Water-hammer signal-processing method.

$$F_{ij} = \sum_{ij} V_i V_j \quad (5)$$

here, V_i represent the cepstrum value with different quefrequency times, and V_j represent the cepstrum value with different physical times at the i -th quefrequency time. The peak position on the curve F_{ij} provide information regarding the reflection of hydraulic fractures.

In the cepstrogram and F_{ij} result, the quefrequency time can represent the reflection time, and the corresponding location in the wellbore is calculated using Eq. (6). Consequently, information regarding hydraulic fractures can be obtained by combining the above with the perforation design. The number and location of perforation clusters within the range of positions obtained through cepstrum analysis are consistent with the number and location of hydraulic fractures, respectively.

$$L = \frac{\tau \times a}{2} \quad (6)$$

where, L is the signal location in the wellbore, τ is the quefrequency, and a is the wave speed. The water-hammer wave speed is calculated as follows [43]:

$$a = \sqrt{\frac{K/\rho}{1 + \frac{K}{E} \frac{D}{e}}} \quad (7)$$

where, K is the bulk modulus of the fluid, ρ the fluid density, E the pipe Young's modulus of elasticity, D the pipe diameter, and e the pipe wall thickness.

2.2.2. Wellbore water-hammer model

Continuity equation of fluid. The continuity equation is expressed as shown in Eq. (8). The mass change of the fluid within a cell over time is equal to the difference between the inflowing and outflowing cell masses, based on the law of mass conservation.

$$A_{pipe} \frac{\partial \rho}{\partial t} + \rho \frac{\partial A_{pipe}}{\partial t} + u A_{pipe} \frac{\partial \rho}{\partial x} + \rho A_{pipe} \frac{\partial u}{\partial x} + \rho u \frac{\partial A_{pipe}}{\partial x} = 0 \quad (8)$$

where A_{pipe} is the cross-sectional area of the fracturing pipe string, t the time, and u the mean velocity in the x -direction.

In an unsteady flow, the size of the fracturing pipe string and the fluid density change with the transient pressure.

$$\begin{cases} \frac{\partial A_{pipe}}{\partial t} = \frac{dA_{pipe}}{dt} - u \frac{\partial A_{pipe}}{\partial x} \\ \frac{\partial \rho}{\partial t} = \frac{d\rho}{dt} - u \frac{\partial \rho}{\partial x} \end{cases} \quad (9)$$

The elastic deformation of the fracturing-pipe string and fluid compressibility are caused by changes in the water-hammer pressure dp . Thus, Eq. (8) can be expressed as

$$\left(\frac{1}{A_{pipe}} \frac{dA_{pipe}}{dp} + \frac{1}{\rho} \frac{d\rho}{dp} \right) \frac{dp}{dt} + \frac{\partial u}{\partial x} = 0 \quad (10)$$

The fluid elasticity term is expressed by the fluid bulk modulus K , and the deformation of the fracturing pipe string can be expressed as a function of the pipe diameter D , wall thickness e , and Young's modulus E_{pipe} using Hooke's law [44,45]:

$$\begin{cases} \frac{1}{\rho} \frac{d\rho}{dp} = \frac{1}{K} \\ \frac{1}{A_{pipe}} \frac{dA_{pipe}}{dp} = \frac{D}{E_{pipe}e} \end{cases} \quad (11)$$

The change ρ in time and space during the water-hammer pressure transmission is insignificant compared with the change in the pressure head H , and ρ is regarded as a constant. Therefore, the continuity equation expressed by the water head can be written as follows:

$$\frac{\partial H}{\partial t} + u \frac{\partial H}{\partial x} + u \sin \theta + \frac{a^2}{g} \frac{\partial u}{\partial x} = 0 \quad (12)$$

here, θ is the angle between the pipe axis and gravitational acceleration g .

Momentum equation of fluid. Fluid flow is characterised by the momentum conservation equation (Navier–Stokes equation), which is expressed as

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} = -\frac{\partial p}{\partial x} + \frac{\partial \tau_{xy}}{\partial x} + \rho g \quad (13)$$

where τ_{xy} denotes the deviatoric stress, which is closely related to flow pressure loss. The friction factor λ of τ_{xy} is typically used to

replace effects caused by fluid viscosity, turbulent disturbances, and pipe characteristics [46,47]. Combining λ with the definition for the pressure head, the final fluid momentum equation is expressed as

$$g \frac{\partial H}{\partial x} + \frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\lambda u |u|}{2D} = 0 \quad (14)$$

here, λ is calculated using the Colebrook–White equation, which is widely used to analyse turbulence in industrial pipes [48]. The Colebrook–White correlation is an implicit equation that must be solved via iterative computation.

$$\frac{1}{\sqrt{\lambda}} = -2 \log_{10} \left(\frac{\delta/D}{3.7} + \frac{2.51}{\text{Re} \sqrt{\lambda}} \right) \quad (15)$$

$$\text{Re} = \frac{\rho \bar{u} D}{\mu} \quad (16)$$

$$\bar{u} = \frac{Q}{A_{\text{pipe}}} \quad (17)$$

where, δ is the wall roughness of the fracturing pipe string, Re is the Reynolds number, and $\bar{u}$ is the average velocity estimated from the flow rate Q .

The fluid within the fracture is affected by the water-hammer wave when it reaches the bottomhole boundary. One can reasonably assume that the reservoir pressure distant from the source remains unaffected by the water-hammer waves owing to the high reservoir capacitance and the fast transient nature of the wave. Consequently, the average near-wellbore pressure $\bar{P}_f$ is utilised to distinguish the pressure head between hydraulic fractures and the bottomhole [15]:

$$\rho g \Delta H = P_{bh} - \bar{P}_f \quad (18)$$

where, P_{bh} is the bottomhole pressure, and $\bar{P}_f$ is determined by fitting an exponential decay curve to the declining portion of the measured surface pressure data, which spans from the instantaneous shut-in pressure (ISIP) to the end of the water-hammer pressure. Notably, the fluids in both the wellbore and fracture must have reached a static state for the surface pressure data to provide accurate measurements. This indicates that the frictional resistance in the wellbore, perforations, and near-wellbore region is eliminated. The steady pressure-decay curve at the end of the water hammer was due to the leak-off of the fracturing fluid. Consequently, a novel method was used in the wellbore water-hammer model to determine the hydraulic fracture pressure.

The shut-in pressure was decomposed into a detrended water-hammer pressure (Eq. (19)) and a steady decaying pressure (Eq. (20)). The latter is represented by an exponential decay equation that allows for the estimation of the ISIP and $\bar{P}_f$ [49].

$$P_{wh} = e^{-\gamma t} M \cos(\omega t - \alpha) \quad (19)$$

$$\bar{P}_f = P_{surf} - P_{wh} = b e^{-mt} + c \quad (20)$$

where, γ is the damping factor of the water hammer, M its magnitude, ω its frequency, α its phase, b is the magnitude of the steady

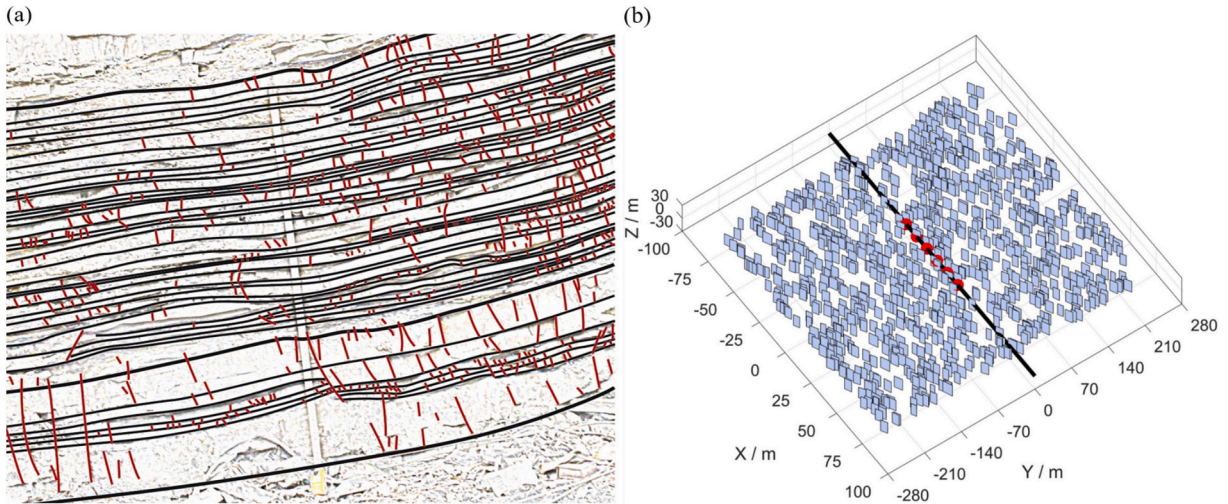


Fig. 4. The distribution of natural fractures in reservoir. (a) Distribution of natural fractures in shale outcrops [54]. (b) Distribution of natural fractures in numerical investigation [39].

exponential pressure decay, m its decay factor, and c its steady-state value. The *ISIP* can be obtained from $b + c$ once shut-in commences.

2.2.3. Natural fracture model

Constructing a natural fracture distribution model is crucial because of the significance of natural fractures in hydraulic fracture propagation and their dimensions within a reservoir. Natural fractures are ubiquitous in reservoir and exhibit irregular distribution characteristics, which can generally be obtained from field geologic data, as shown in Fig. 4. The red and blue lines represent natural fractures in shale outcrops and geomaterial, respectively. In this model, a natural fracture is regarded as a vertical plane with a constant height. Assuming that the midpoint of the natural fractures adheres to a uniform random distribution, the number of natural fractures N_{nf} in a specified area can be calculated as follows:

$$N_{nf} = \rho_{nf} A_{re} \quad (21)$$

here, ρ_{nf} is the density of natural fractures (number/m²) and A_{re} is the area of the reservoir. The natural fracture width W_{nf} and length L_{nf} can be described using a normal distribution [39,50]:

$$W_{nf} = N(W_{aver}, \sigma_W^2) \quad (22)$$

$$I = \frac{4\rho L_f}{\pi W_f H_f} = \frac{32\rho L_f}{\pi^3 P_{net} H_f^2} \frac{E_{rock}}{1 - \nu^2} E(x) \quad (23)$$

here, W_{aver} is the average width of natural fractures, σ_W^2 is the standard deviation of the width of natural fractures, L_{aver} is the average length of natural fractures, and σ_L^2 is the standard deviation of the length of natural fractures. The natural fracture width and length are typically 0–100 μ m and 0–50 m, respectively [51,52].

The closure of hydraulic fractures, which reduces the fracture width, is primarily caused by the fracturing fluid leaking off into the reservoir. The fracturing fluid leak-off model is expressed as shown in Eq. (24). Carter's model determines the leak-off into the reservoir matrix, and the leak-off into the natural fracture is calculated based on the reservoir pore pressure [53]. The total amount of leak-off from positive and negative water hammer is assumed to be zero in this case, and the fracturing fluid leak-off is due to the average pressure (the average near-wellbore pressure $\bar{P}_f$) in the hydraulic fracture.

$$V_{leak} = \int_{t_0}^t 4C_{leak} H_f L_f / (t - t_{open})^{1-\psi} dt + 2L_f L_{nf} \cos\theta_{nf} \rho_{nf} \int_{t_0}^t \frac{A_{nf} W_{nf}^2 (\bar{P}_f(t) - P_{pore})}{12\mu L_{nf}} dt \quad (24)$$

here, V_{leak} is the leak-off total volume (m³), t_0 is the shut-in time (s), C_{leak} is the leak-off coefficient, H_f is the height of hydraulic fracture (m), L_f is the half-length of hydraulic fracture (m), θ_{nf} is the strike angle of natural fracture, which will be presented more comprehensively in the following section, ψ is the liquid leak-off index ($\psi = 0.5$ for Newtonian fluids), P_{pore} is the pore pressure of the reservoir (Pa). t_{open} is the opening time of hydraulic fracture (s), which is defined as the time which the fracture wall is in contact with the fracturing fluid and is set as the moment when the formation begins to fracture. A_{nf} is the cross-sectional area of natural fracture, which is estimated based on the width and height of natural fracture (m²). The two items on the right-hand side denote the leak-off volume from the hydraulic fracture into the reservoir matrix and natural fracture, separately. The leak-off area into the reservoir matrix was an ellipse with long-axis $2L_f$ and short-axis H_f . The leak-off into a natural fracture depends on the natural fracture distribution and differential pressure. If the average pressure of the hydraulic fracture exceeds the opening pressure of the natural fracture, then the natural fracture is closed and the second term becomes zero.

Furthermore, the strike angle between the hydraulic and natural fractures significantly affects the hydraulic fracture propagation and dimensions. There are several interaction patterns between hydraulic fracture and natural fracture, such as net-crossing, net-opening, crossing-opening, and opening-crossing [55,56]. In this study, we focused on the two most representative patterns—net-crossing and net-opening, because they effectively capture the characteristics of fluid pressure dissipation. Under identical conditions

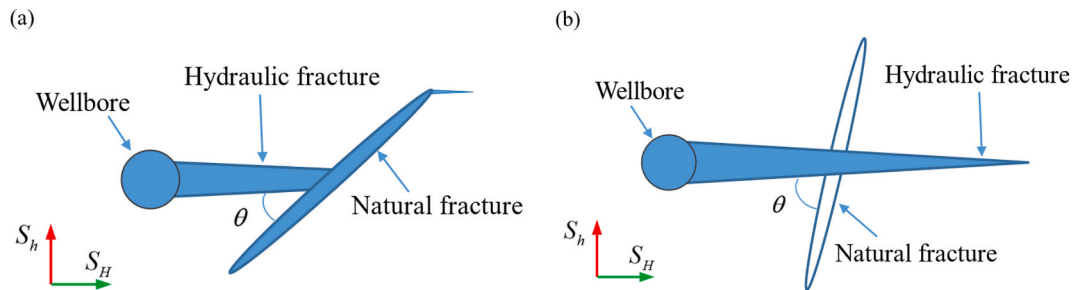


Fig. 5. Schematic illustration of effect of natural fracture strike angle on hydraulic fracture propagation in vertical well: (a) Opening model, (b) Crossing model.

(horizontal stress difference, fluid viscosity, fluid rate, coefficient of interfacial friction, etc.), the hydraulic fracture propagation mode changes from opening to crossing as the strike angle increases, as shown in Fig. 5. The length of the overall fracture increases because of less fracturing fluid and a lower pressure drop propagate the natural fracture. The propagation direction of the hydraulic fracture is along the maximum horizontal stress, S_H . The criterion for hydraulic fractures crossing natural fractures based on the results of [57] was employed, where the coupled effect of the fluid flow, stress shadow, and poroelastic response was considered. The determination of whether a hydraulic fracture crosses or opens a natural fracture is based on evaluating the strike angle under varying mechanical properties. Hydraulic fractures are more likely to cross natural fractures with larger strike angles and greater horizontal stress differences. The activated natural fracture aperture was assumed to be uniform in space and similar to the fracture width.

In 2D space, the Fisher distribution can be used to describe the strike angle between hydraulic and natural fractures:

$$\theta_{nf} = \theta_{aver} + \arccos\left(\frac{\ln(1 - K_{rand})}{K_\theta} + 1\right) \quad (25)$$

where θ_{nf} is the strike angle of the natural fracture, which is between 0° and 90° , θ_{aver} is the average strike angle of natural fracture, K_θ is the deviation coefficient, and K_{rand} is a random number between 0 and 1.

The fracturing fluid flow inside a fracture is generally simplified as a lubrication flow between two smooth plates [58]. Although the opening of natural fractures results in the formation of complex fracture networks, it also decreases the pressure owing to the bending of the flow line, as shown in Fig. 5(a). Eq. (26) was formulated to estimate the additional natural fracture pressure drop ΔP_{nf} based on branch flows with different junctions [59,60]. The distribution of natural fractures on the hydraulic fracture propagation plane is consistent before and after deflection.

$$\Delta P_{nf} = \frac{1}{2}\rho\left(\frac{2Q_0}{\pi H_f W_f}\right)^2 \left[\left(\frac{W_f}{W_{nf}}\right)^2 + 1 - 2\frac{W_f}{W_{nf}}\cos\left(\frac{3}{4}\theta_{nf}\right) \right] \times 2L_f L_{nf} \cos\theta_{nf} \rho_{nf} \quad (26)$$

here, W_f is the width of hydraulic fracture (m) and Q_0 is the steady-state flow rate prior to fracturing shut-in (m^3/s). Eq. (26) depends on the hydraulic fracture dimensions. Therefore, an iterative calculation between the natural fracture pressure drops and the hydraulic fracture dimensions is required.

2.2.4. Hydraulic fracture dimension estimation model

RCI multifracture model. The RCI model is vital to the coupling of hydraulic fractures into the water-hammer equation and has been widely adopted and validated. The transient pressure–flowrate behaviour of any hydraulic fracture is governed by its fluid and volume, which is reflected through the R , C , and I [15]. The hydraulic fracture behaviour is presented through a series combination in the form shown in Eq. (27). As the fluid enters the fracture up to the maximum compliance, additional fluid can only enter under a net pressure increase. This is analogous to charging a capacitor in a series circuit.

$$\rho g \Delta H = RQ + \frac{1}{C} \int Q dt + I \frac{dQ}{dt} \quad (27)$$

In the equation above, the three terms on the right-hand side denote the changes in pressure due to frictional resistance, the volume change in the fracture, and the fluid inertia, respectively. The resistance R is defined as the constant of proportionality between the flow rate Q to maintain the fracture and the required potential energy difference [7]. This can be determined by calculating the near-wellbore friction drop ΔP_{nwf} (including the wellbore perforation friction and fracture bending friction) and the steady-state flow rate Q_0 before fracturing shut-in, as follows:

$$R = \frac{\Delta P_{nwf}}{Q_0} \quad (28)$$

The capacitance C is equivalent to the fracture compliance, which characterises the rock strain under external forces and reflects the relationship between the fracture's dimensional parameters and pressure behaviour. C is defined as the ratio of the liquid volume stored ΔV in a system to the change in the liquid pressure.

$$C = \frac{\Delta V}{\Delta P} \quad (29)$$

The inductance I is related to the fluid mass in the system and is the pressure difference required to accelerate or decelerate the flow rate in the absence of friction. It is equivalent to the inductance in the circuit [61], which can be expressed as:

$$I = \frac{\Delta P \cdot dt}{dQ} \quad (30)$$

The simulated water-hammer pressure signals were matched to the surface data by iteratively adjusting the values of R , C , and I . The resulting values can be used to estimate the subsequent hydraulic fracture dimensions. The systematic method for matching the surface data is as follows [62]: First, R was iteratively adjusted until the simulated pressure signal matched the amplitude and decay of the field data. Subsequently, C was adjusted until the simulated pressure signal matched the period and mean pressure of the field data. Finally, I was adjusted iteratively until the calculated hydraulic fracture height was similar to the reservoir thickness targeted for the fracturing treatment.

Fracture dimension estimation model. The average net pressure $\overline{P_{net}}$ is defined as the difference between the average pressure inside the fracture ($\overline{P_f} = ISIP - \Delta P_{nwf}$) and the formation closure pressure typically substituted by the horizontal minimum stress S_h . It is a crucial parameter for computing the length, width, and height of hydraulic fractures [63]. The net pressure exerts directly on the rock to create hydraulic fractures and represents a portion of the effective pressure involved in the process. The average net pressure $\overline{P_{net}}$ can be obtained using Eq. (31), after determining the respective R , C , and I values [62]. The near-wellbore friction drop ΔP_{nwf} can be obtained using Eq. (28).

$$\overline{P_{net}} = (ISIP - \Delta P_{nwf}) - S_h \quad (31)$$

Multiple hydraulic fractures are created simultaneously during the multicluster fracturing treatment in unconventional reservoirs to enhance the stimulated region. Strain perturbations occur near several induced hydraulic fractures owing to their interference, thus resulting in the mechanical-stress shadow effect, which affects the dimensions of the induced hydraulic fractures and results in non-uniform propagation. Due to the complex and uncertain distribution of natural fractures within the reservoir, the stress interference effects caused by natural fractures are not considered in the present model. Hydraulic fracturing in naturally fractured reservoir, particularly in shales where horizontal wells are drilled, shows that fractures typically propagate laterally rather than vertically [64]. The hydraulic fracture length typically exceeds its height. The stress shadow area is determined based on the minimum size of the fractured surface. Consequently, the effect of stress shadow is dominated by the hydraulic fracture height [65,66]. As shown in Fig. 6, the stress shadow ΔP_{shadow} for a semi-infinite hydraulic fracture can be expressed as [49]:

$$\Delta P_{shadow} = P_{net} \frac{2r}{h_f} \left(\frac{h_f^2}{4r_1 r_2} \right)^{\frac{3}{2}} \sin \theta \sin \frac{3}{2} (\theta_1 + \theta_2) - P_{net} \left[\frac{r}{(r_1 r_2)^{1/2}} \cos \left(\theta - \frac{1}{2} \theta_1 - \frac{1}{2} \theta_2 \right) - 1 \right] \quad (32)$$

where, P_{net} is the net pressure of the hydraulic fracture, which is the source of the induced stress shadow, r is the distance between the centre of the fracture and point M , r_1 and r_2 are the distances between the right and left ends of the fracture with point M , respectively, and θ is the angle about the fracture surface and the attachment between point M and the centre of the fracture. Specifically, θ_1 and θ_2 are the angles about the fracture surface and the attachment between point M and the right and left ends of the fracture, respectively. To simplify the problem, the stress shadow between the hydraulic fractures was equated to that at a position on a line perpendicular to the fracture plane and passing through its centre (the x -axis in Fig. 6). Finally, the stress shadow generated by multifracture was obtained using the superposition principle.

Therefore, a new average net pressure can be obtained by correcting Eq. (31) while, considering the multifracture stress shadow

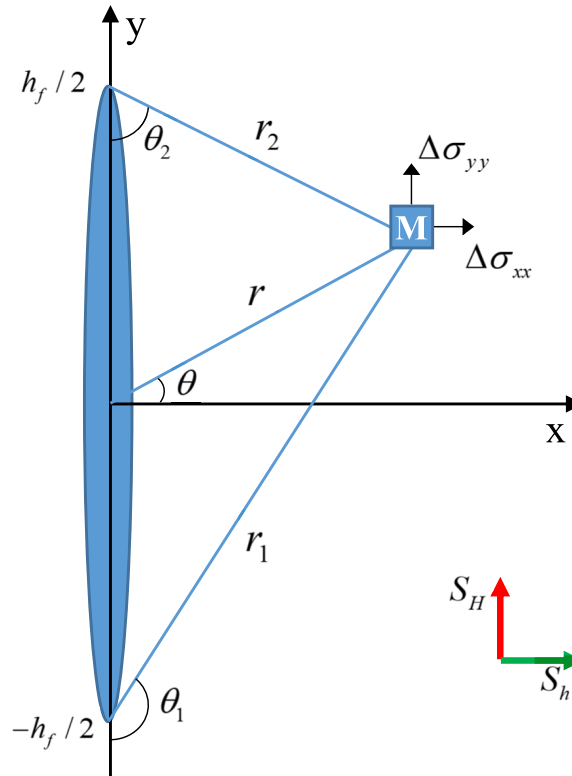


Fig. 6. Schematic illustration of stress shadow on point M caused by hydraulic fracture.

(Eq. (32)) and natural fracture pressure drop (Eq. (26)).

$$\overline{P_{net}} = (ISIP - \Delta P_{nwf}) - S_h - \sum_{n=1}^N \Delta P_{shadow}(n) - \sum_{n=1}^N \Delta P_{nf}(n) \quad (33)$$

The multifracture system can be modelled using a new RCI circuit, in which the capacitance of each fracture is connected in parallel and thus shares the same R and I . The multifracture equivalent circuit relationship is shown in Fig. 7. A multifracture system is equivalent to a bi-wing fracture system. Subsequent to the fracturing shut-in, the multifracture system connects to the wellbore, thus affecting the water-hammer signal. The length and width of an equivalent hydraulic fracture (E-frac) can be determined by combining all hydraulic fractures connected to a wellbore [7]. The capacitance of each hydraulic fracture is connected in parallel in the RCI circuit, thus allowing the total capacitance of all hydraulic fractures to be expressed as

$$C = \sum_{n=1}^N C_n \quad (34)$$

where C is the total capacitance and C_n is the capacitance of the n -th hydraulic fracture.

In a previously conducted hydraulic fracturing energy-analysis study [67], a method was devised for determining the relationship among the dimensions of a lengthy fracture, the average net pressure, and the rock properties. The equation proposed was as follows:

$$\overline{P_{net}} = \frac{4}{\pi^2} \frac{E_{rock}}{1 - \nu^2} E(m) W_f \frac{2}{H_f} \quad (35)$$

where E_{rock} is the Young's modulus of rock and ν is the Poisson's ratio of rock, $E(m)$ is a complete elliptical integral of the second type and is related to the hydraulic fracture dimension. It can be calculated as:

$$E(m) = \int_0^1 (1 - t^2)^{-\frac{1}{2}} (1 - mt^2)^{\frac{1}{2}} dt \quad (36)$$

$$m = 1 - \left(\frac{H_f}{2L_f} \right)^2 \quad (37)$$

Because the change in the fracture capacitance is caused only by the change in the width, the capacitance defined in Eq. (29) can be rewritten by considering the leak-off volume in Eq. (24):

$$C = \frac{\pi H_f L_f \Delta W_f}{2\Delta P} + \frac{4H_f L_f \int_{t_0}^t C_{leak} / (t - t_{open})^{1-\psi} dt}{ISIP - P_{end}} + \frac{2L_f L_{nf} \cos \theta_{nf} \rho_{nf} \int_{t_0}^t A_{nf} W_{nf}^2 (\overline{P}_f(t) - P_{pore}) dt}{12\mu L_{nf} (ISIP - P_{end})} \quad (38)$$

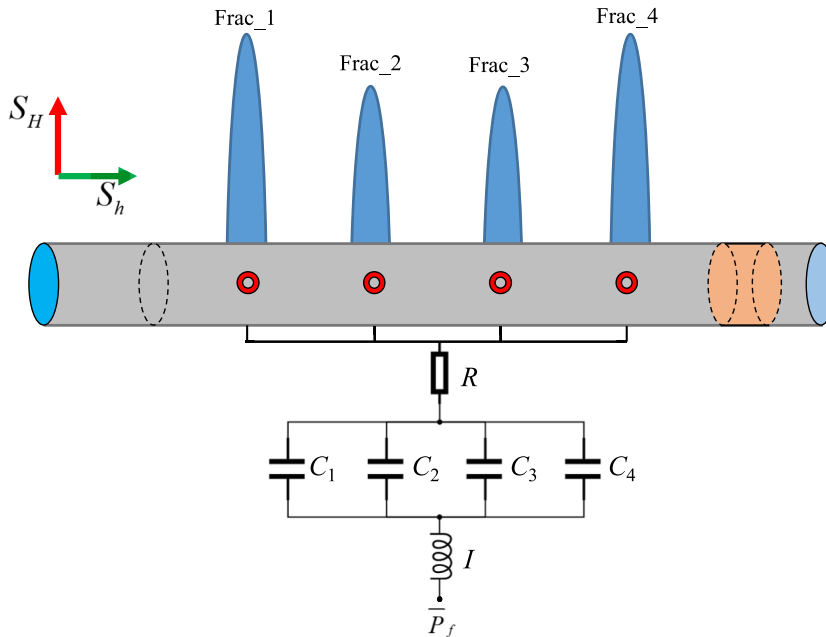


Fig. 7. Schematic of a multifracture RCI circuit.

where, the three terms on the right-hand side of the equation represent the fracture system compliance owing to rock strain, reservoir matrix leak-off, and natural fracture leak-off, respectively. P_{end} is the pressure at the end of the water hammer, which can be obtained from the surface pressure data. Combining the equation above with Eq. (35), the final expression for the relationship between the parameter C and the hydraulic fracture dimensions can be written as follows:

$$C = \frac{\pi^3 H_f^2 L_f (1 - \nu^2)}{16 E_{rock} E(m)} + \frac{4 H_f L_f \int_{t_0}^t C_{leak} / (t - t_{open})^{1-\psi} dt}{ISIP - P_{end}} + \frac{2 L_f L_{nf} \cos \theta_{nf} \rho_{nf} \int_{t_0}^t A_{nf} W_{nf}^2 (\bar{P}_f(t) - P_{pore}) dt}{12 \mu L_{nf} (ISIP - P_{end})} \quad (39)$$

Based on Newton's second law, the I in Eq. (30) can be expressed in terms of the dimensions of the ellipsoidal hydraulic fracture as follows:

$$I = \frac{4 \rho L_f}{\pi W_f H_f} = \frac{16 \rho L_f}{\pi^3 P_{net} H_f^2} \frac{E_{rock}}{1 - \nu^2} E(m) \quad (40)$$

The number and location of hydraulic fractures, the distribution of natural fractures, the $ISIP$, and the near-wellbore pressure $\bar{P}_f(t)$ can be determined based on field geological data and surface pressure signals. Meanwhile, the values of R , C , and I can be obtained by matching the simulated water-hammer pressure with the surface pressure signal. Therefore, the average net pressure $\bar{P}_{net}$ of the E-frac can be calculated using Eqs. (28) and (33). Finally, the three unknown quantities, i.e. the half-length L_f , width W_f , and height H_f , of E-frac can be obtained by iteratively solving Eqs. (39) and (40). The E-frac height H_f can be regarded as equivalent to the height of each propagated hydraulic fracture, as follows:

$$H_f = H_{f,n} (n = 1, 2, \dots, N) \quad (41)$$

The determined length and width of the E-frac are a combination of all the propagated hydraulic fractures in communication with the wellbore. Based on the abovementioned multifracture RCI circuit assumption, the equivalent single-fracture width and half-length can be written as the sum of each propagated hydraulic fracture [7], and the conversion equations are as follows:

$$W_f = \sqrt{\sum_{n=1}^N W_{f,n}^2} \quad (42)$$

$$L_f = \sqrt{\sum_{n=1}^N L_{f,n}^2} \quad (43)$$

where $W_{f,n}$ (m) and $L_{f,n}$ (m) denote the width and length of the n -th propagated hydraulic fracture, respectively.

The width of each hydraulic fracture is calculated based on its average net pressure using Eqs. (42) and (44), as the fluid pressure within the fracture determines the width by resisting the formation stress [22]. The length of each hydraulic fracture is assumed to be known, as it is required for calculating the average net pressure using Eq. (44). Once the value of $W_{f,n}$ is established, it can be substituted into Eq. (45) to obtain the updated value of $L_{f,n}$. Thus, $W_{f,n}$ and $L_{f,n}$ can be updated repeatedly until an acceptable residual is achieved. This method is advantageous as it considers the interaction between the lengths and widths of multifracture. In this method,

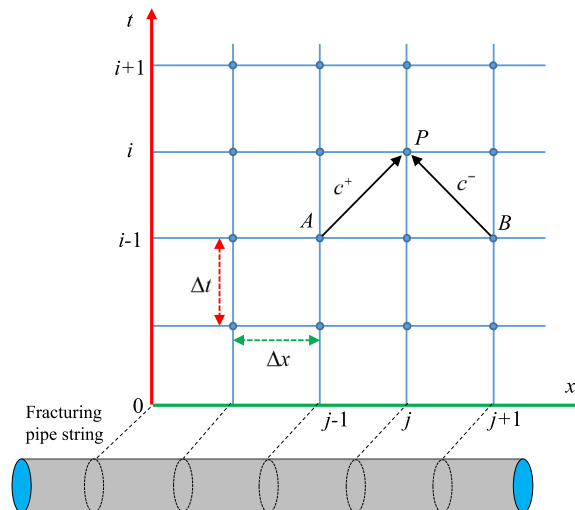


Fig. 8. Mesh for water-hammer calculation based on MOC.

establishing a larger initial length and width for the hydraulic fracture near the wellbore root is reasonable to account for the friction within the horizontal wellbore. This is supported by field diagnostic experience and numerical simulation results [68,69].

$$\overline{P_{net}}(n) = (ISIP - \Delta P_{wvf}) - S_h - \Delta P_{shadow}(n) - \Delta P_{nf}(n) \quad (44)$$

$$L_{f,n} = \frac{16\rho}{I\pi^3 \overline{P_{net}}(n) H_{f,n}^2} \frac{E_{rock}}{1 - \nu^2} E(m)_n \quad (45)$$

2.3. Model solution

2.3.1. Method of characteristics

The main numerical solution methods for the water-hammer model include the method of characteristics (MOC), finite-difference calculus, and finite-volume method [70,71]. The explicit MOC is the most widely used method for simulating water-hammer flows owing to its simplicity, accuracy, and coding efficiency. When using the MOC, the mesh of the calculated time and spatial domains was segmented, as shown in Fig. 8.

The wellbore flow-continuity and momentum equations were rewritten as expressions for the pump rate to better characterise the parameters used in the fracturing construction, as follows:

$$L_1 : A \frac{\partial H}{\partial t} + Q \frac{\partial H}{\partial x} + Q \sin \theta + \frac{a^2}{g} \frac{\partial Q}{\partial x} = 0 \quad (46)$$

$$L_2 : gA \frac{\partial H}{\partial x} + \frac{\partial Q}{\partial t} + \frac{Q}{A} \frac{\partial Q}{\partial x} + \frac{\lambda Q |Q|}{2DA} = 0 \quad (47)$$

The partial differential equations above can be transformed into ordinary differential equations using the MOC, which defines a parameter χ and combines it linearly as $\chi L_1 + L_2 = 0$:

$$\left[\frac{\partial Q}{\partial t} + \frac{\partial Q}{\partial x} \left(\omega \frac{a^2}{g} + \frac{Q}{A} \right) \right] + \left[\chi A \left(\frac{\partial H}{\partial t} + \frac{\partial H}{\partial x} \frac{\omega Q + gA}{\omega A} \right) \right] + \chi Q \sin \theta + \frac{\lambda Q |Q|}{2DA} = 0 \quad (48)$$

By comparing the two left terms in the equation, i.e. the pressure head H and flow rate Q material derivative, the parameter values χ , and the relationship between the spatial mesh Δx and temporal mesh Δt can be obtained as follows:

$$\begin{cases} \omega = \pm \frac{g}{a} \\ \frac{\Delta x}{\Delta t} = \pm a \end{cases} \quad (49)$$

Substituting these values into Eq. (48) yields the two characteristic equations expressed in Eqs. (50) and (51). Therefore, the ordinary differential equations can only be solved based on the characteristic line c^+ or c^- , as shown in Fig. 8. This implies that the water-hammer parameters (H and Q) at the next moment are determined by both the upstream and downstream nodes at the final moment. The intersection P of the two characteristic lines is calculated using points A and B in Fig. 8. Therefore, the flow at the current moment can be used to calculate the head pressure and flow rate at any point later in time as follows:

$$c^+ : \frac{dQ}{dt} + \frac{gA}{a} \frac{dH}{dt} + \frac{gQ}{a} \sin \theta + \frac{\lambda Q |Q|}{2DA} = 0 \left(\frac{dx}{dt} = a \right) \quad (50)$$

$$c^- : \frac{dQ}{dt} - \frac{gA}{a} \frac{dH}{dt} - \frac{gQ}{a} \sin \theta + \frac{\lambda Q |Q|}{2DA} = 0 \left(\frac{dx}{dt} = -a \right) \quad (51)$$

Further finite-difference discretisation of the c^+ and c^- subtraction equations yields the following equation:

$$c^+ : H(i, j) = H(i-1, j-1) + \frac{a \Delta t}{gA} \left(-\frac{Q(i, j) - Q(i-1, j-1)}{\Delta t} - \frac{g \sin \theta}{2a} Q(i-1, j-1) - \frac{g \sin \theta}{2a} Q(i, j) - \frac{\lambda^+ |Q(i-1, j-1)|}{2DA} Q(i, j) \right) \quad (52)$$

$$c^- : H(i, j) = H(i-1, j+1) + \frac{a \Delta t}{gA} \left(\frac{Q(i, j) - Q(i-1, j+1)}{\Delta t} - \frac{g \sin \theta}{2a} Q(i-1, j+1) - \frac{g \sin \theta}{2a} Q(i, j) + \frac{\lambda^- |Q(i-1, j+1)|}{2DA} Q(i, j) \right) \quad (53)$$

here, i and $i+1$ represent the present time and the next time step, respectively. j and $j+1$ represent the present position and the right position node, respectively. λ^+ and λ^- are the flow friction factors calculated based on characteristic lines c^+ and c^- , respectively. The Reynolds number is calculated using the flow rate of the final time step.

2.3.2. Boundary conditions

Fig. 8 shows that only one characteristic equation is available for both the wellhead and bottomhole fracture. Boundary conditions

are required to define the head pressure, flow rate values, or the relationship between them. The wellhead condition can be expressed by the relationship between the flow rate and time after fracturing shut-in, where the flow rate is assumed to decrease continuously, as expressed in Eq. (54). This can be obtained using a wellhead flow-rate sensor. The shut-in time is the time required for the steady-state flow rate to decrease to zero. The wellhead value can be calculated directly from the characteristic line equation c^- . The initial conditions are set based on the steady-state flow status before the fracturing shut-in.

$$Q(i, 1) = Q_0(t) \quad (54)$$

For the wellbore boundary conditions, the *ISIP* and near-wellbore pressure are first determined by fitting the surface pressure data using the method described in Section 2.2.2. Subsequently, the pressure head difference between the wellbore node and the fracture is calculated using Eq. (27). The flow rate values at the wellbore can be obtained using the joint characteristic line equation, i.e. the c^+ solution. Finally, various *RCI* values are set to match the field water-hammer pressure signal.

2.4. Model validation

The diagnostic model proposed in this study primarily comprises three components: number and location analyses of hydraulic fractures, water-hammer pressure simulation, and estimation of hydraulic fracture dimension. The feasibility of each component must be verified independently because they are separated from each other in the calculations. Table 1 lists the basic parameters used for the model-validation calculations.

First, the ability of the diagnostic model to identify both the number and location of hydraulic fractures was evaluated by analysing artificial water-hammer signals generated by simulating a water hammer at a constant pressure of 50 MPa along the fracture boundaries, while incorporating varying hydraulic fracture information. The values of R , C , and I were fixed at $1 \times 10^7 \text{ Pa} \cdot \text{s}/\text{m}^3$, $1 \times 10^{-5} \text{ m}^3/\text{Pa}$, and $5 \times 10^6 \text{ Pa}/\text{m}^3 \cdot \text{s}^{-2}$, respectively. Table 2 provides the details of the hydraulic fracture settings used in the analysis. Hydraulic fractures were artificially set at different locations, numbers, and spacing. A comprehensive analysis of each case is presented in Appendix B. A quantitative analysis was performed using the relative deviations of the peak locations within the half-peak range. Although the resolution decreased as the fracture spacing increased, all cases indicated relative deviations of less than 10 m, and the position of the cepstrum peak accurately indicated the fracture location. This study demonstrates the ability to detect the number and location of hydraulic fractures.

In addition, the water-hammer pressure simulation model was validated using the computational fluid dynamics software Fluent in accordance with the validation method outlined in previous studies [22]. Fluent is widely accepted as a reliable numerical simulation software for investigating engineering pipe flows [72]. Water hammer was simulated in pipe flow based on a constant-pressure outlet boundary. To ensure an efficient operation, we established a simplified pipeline with a length of 5 m, a diameter of 0.2 m, a stabilised flow rate of $2 \text{ m}^3/\text{min}$, and an outlet pressure of 20 MPa. A numerical calculation case of water hammer for a pure water flow rate decreasing to zero in 0.002 s. The comparative results of the water-hammer calculations are presented in Fig. 9. Our water-hammer model yielded that agreed well with the Fluent results in terms of the amplitude, period, and decay, with a maximum error of 0.26 MPa. The unsatisfactory agreement at the onset of water hammer was due to the low stability of the Fluent transient calculation.

Finally, the feasibility of the hydraulic fracture dimension estimation model was analysed by comparing with the fracture dimension results obtained from Carey's field case studies. The R , C , and I for this process were known, whereas the fracture dimensions were estimated directly. Fracture length, fracture width, and fracture height change between 70–210 m, 27–53 mm, and 16–30 m, respectively. More detailed information on the computational case can be found in study of [7]. The relative error results are shown in Fig. 10. As shown, the model's fracture length, width, and height were consistent with those of Carey's, with a relative error of less than 5 %. Thus, the hydraulic fracture dimension-estimation model discussed herein can be considered reliable.

3. Results and discussion

3.1. Water-hammer pressure analysis

Water-hammer pressure pulsations dissipated in the wellbore and fractures after the fracturing treatment shut-in. These pulsations are beneficial as they contain hydraulic fracture information for fracturing diagnostics, although they jeopardise the wellbore integrity (e.g. casing deformation and sand production). Therefore, it is necessary to analyse the water-hammer pressure characteristics must be analysed. In this study, the water-hammer pressures were numerically calculated for different shut-in times (0.1–4.0 s) and

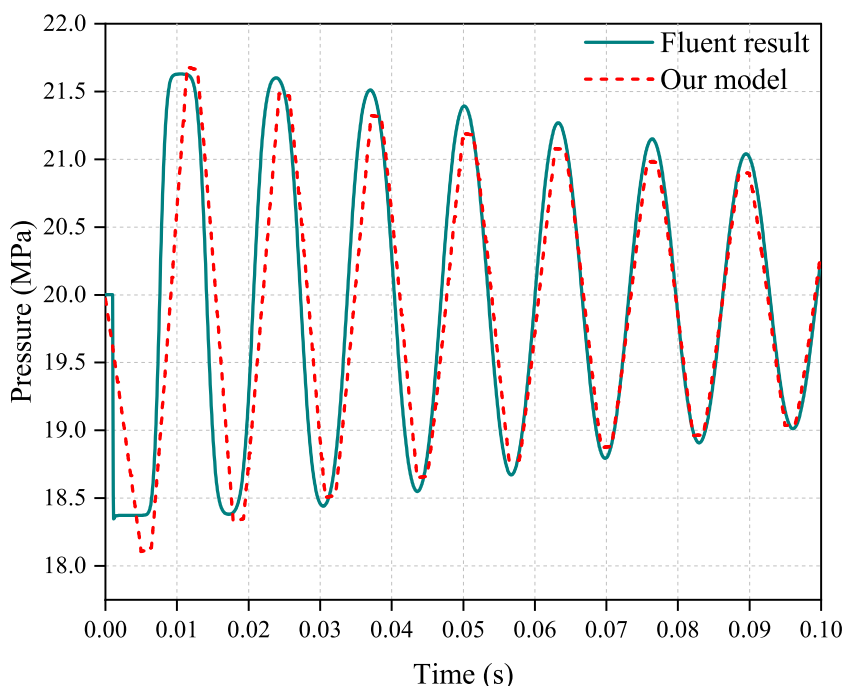
Table 1
Basic parameters for calculations.

Parameter	Value
Well depth	2000 m
Pipe size	124.26 mm
Steady flow rate	15 m ³ /min
Fluid viscosity	1000 kg/m ³
Fluid density	5.0 mPa · s
Speed of water hammer wave	1000 m/s

Table 2

Settings of artificial hydraulic fractures for number and location analyses of hydraulic fractures.

Parameter	Case1	Case2	Case3	Case4	Case5	Case6	Case7
Location of fractures (m)	1800	1800 1700	1800 1750	1800 1790	1800 1700 1600	1800 1700 1600 1500	1800 1700 1600 1500 1400
Number of fractures (number)	1	2	2	2	3	4	5
Spacing of fractures (m)	/	100	50	10	100	100	100

**Fig. 9.** Comparison results of water hammer calculation with fluent software.

displacements (1–20 m³/min). The shut-in time and displacement settings were derived from typical field fracturing and pump shutdown procedures. The other input parameters are listed in Table 1. The wellhead pressure initially decreased after the fracturing shut-in. Therefore, the envelope of the minimum peak was adopted to characterise the water-hammer pressure, and the results are presented in Fig. 11. The amplitude of the water-hammer pulsation decreased and stabilised with time under the effects of flow resistance and formation-pressure supplementation. The strongest pressure pulsations occurred at the onset of water hammer owing to the conversion between flow momentum and pressure. As shown in Fig. 11(a), the pressure pulsation increased as the shut-in time decreased during the initial stage of the water hammer. The reduction in the shut-in time from 4 to 0.1 s decreased the strongest pressure pulsation by 4.2 MPa. This reduction was caused by the faster transformation of the flow momentum to pressure at a fixed displacement under a reduced shut-in time. Fig. 11(b) shows the effect of displacement on the water-hammer pressure pulsation, which indicates that the pressure pulsation is stronger at higher displacements. The displacement exerted a greater effect on the variations in the water-hammer pressure than the shut-in time. The strongest pressure pulsation reduced by 19.8 MPa when the displacement was reduced from 20 to 1 m³/min. Furthermore, a higher-pressure decay rate (stabilisation rate) was achieved under a higher displacement. This is because a greater displacement implies more momentum conversion and greater flow resistance to pressure decay. Therefore, to avoid significant pressure pulsations that can damage the wellbore integrity during fracturing shut-in, a staged pump-off can be used along with an extended shut-in time to prevent a considerable decrease in the displacement. Similarly, the fracturing shut-in method can be modified accordingly to obtain a clear water-hammer signal for subsequent fracturing diagnostics.

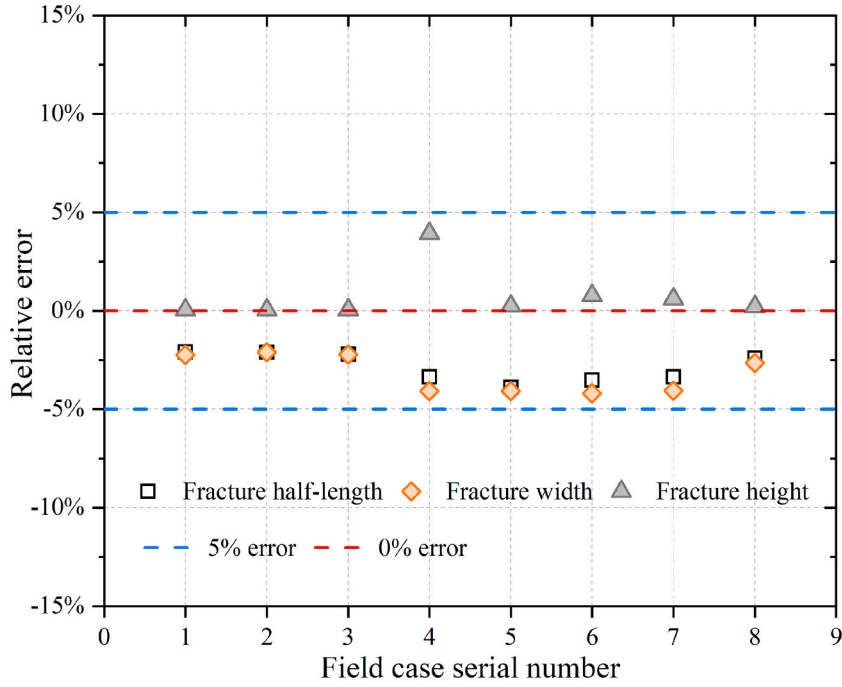


Fig. 10. Relative error results of hydraulic fracture dimension estimation.

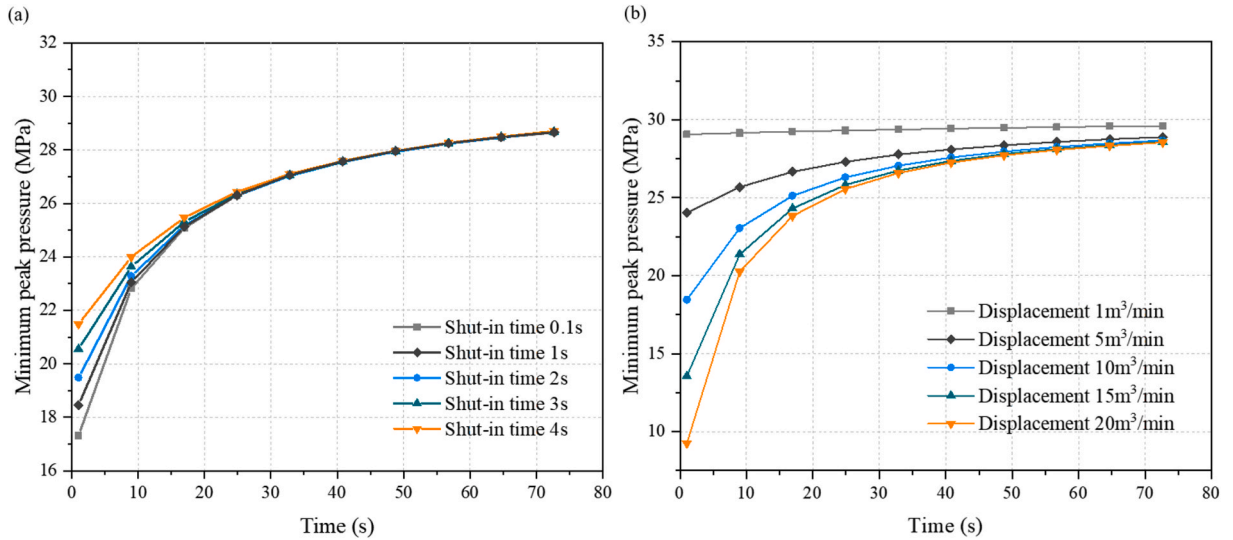


Fig. 11. Envelope of minimum water-hammer pressure peak for different displacements and shut-in times: (a) Fixed displacement of $10 \text{ m}^3/\text{min}$, (b) Fixed shut-in time of 1 s .

3.2. Hydraulic fracture dimension analysis

The design of fracturing treatment and geology significantly affect the propagation of multifracture. In this section, the *RCI* model is used to analyse multifracture dimensions in naturally fractured reservoir with different perforation cluster designs, natural fracture densities, and horizontal stresses. Table 3 lists the calculation conditions, which are based on typical perforation designs and naturally fractured shale from staged multi-cluster hydraulic fracturing practices [39,73]. The values of *R*, *C* and *I* are fixed in these cases.

3.2.1. Perforation cluster design

Perforation design is a vital optimisation method in multicluster fracturing for enhancing the stimulated reservoir volume and exploiting subsurface resources. The effect of perforation design on multifracture dimensions must not be disregarded. In this section,

Table 3
Calculation conditions for hydraulic fracture dimension analysis.

Parameter	Base value	Range of change
R	$1.3 \times 10^7 \text{ Pa} \cdot \text{s}/\text{m}^3$	Fixed value
C	$4.0 \times 10^{-5} \text{ m}^3/\text{Pa}$	Fixed value
I	$3.5 \times 10^6 \text{ Pa}/\text{m}^3 \cdot \text{s}^{-2}$	Fixed value
Cluster number	4	1–6
Cluster spacing	15 m	5–25 m
Density of the natural fracture	0.1 number/ m^2	0–0.2 number/ m^2
Minimum horizontal stress	50 MPa	45–60 MPa
Horizontal stress difference	10 MPa	5–20 MPa

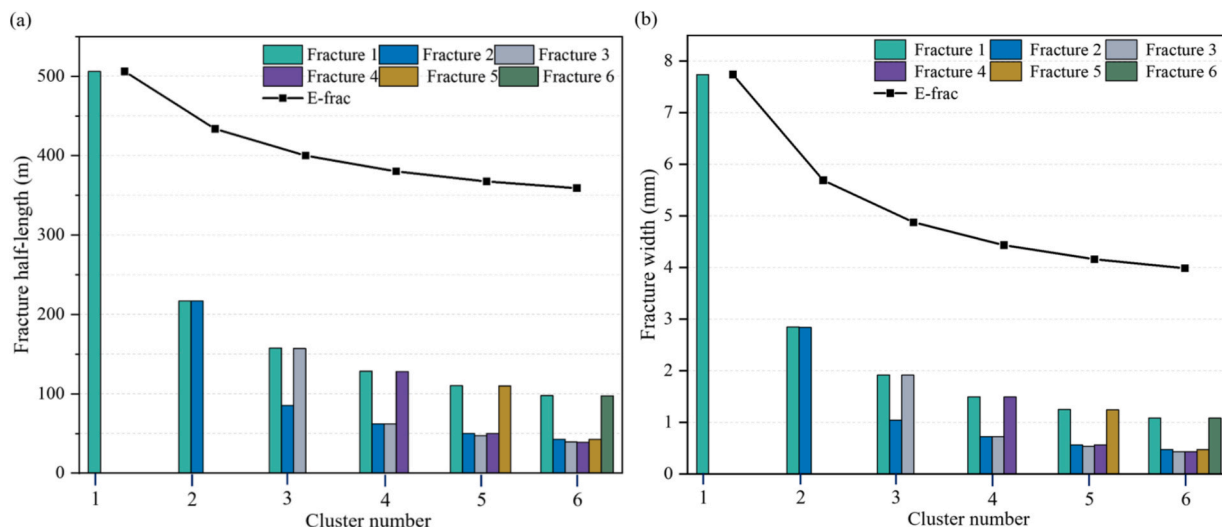


Fig. 12. Fracture dimension under different number of perforation clusters: (a) Fracture half-length, (b) Fracture width.

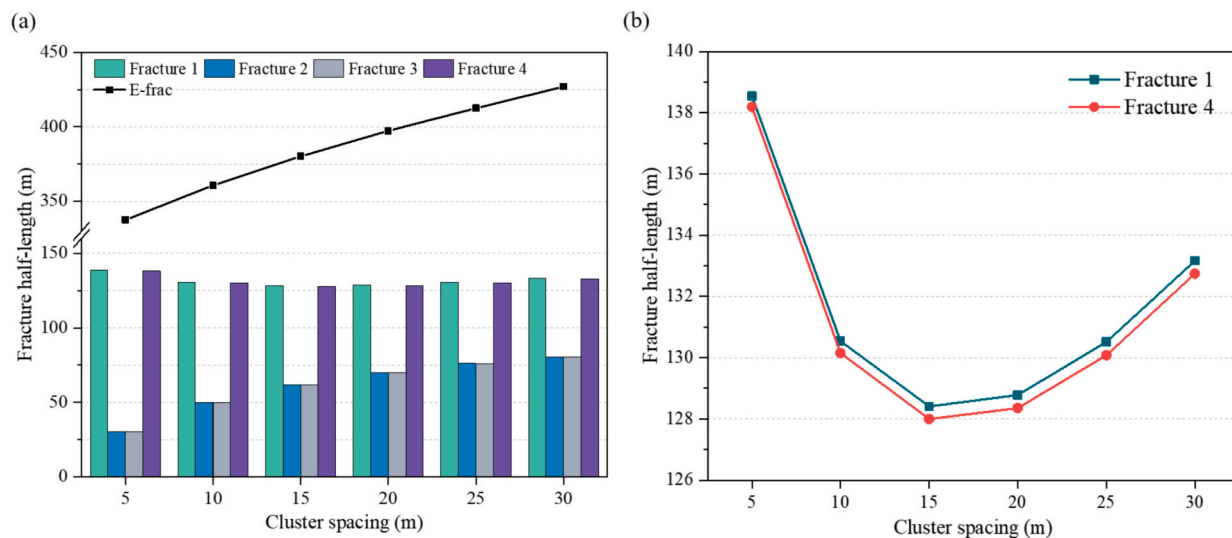


Fig. 13. Fracture half-length under different perforation cluster spacings: (a) E-frac and multifracture assemblies, (b) First and final fracture of multifracture.

we provide an analysis of fracture diagnosis based on different perforation cluster numbers (1–6) and spacings (5–25 m).

(1) Perforation cluster number

Fig. 12 presents the half-length and width-estimation results of the E-frac and multifracture for different numbers of perforation clusters. The fracture number i denotes their spatial location order, which implies that they are the i -th hydraulic fracture near the heel end of the fracturing stage. The height of the fracture is established by the reservoir thickness within the model used in this study and is thus excluded from the present discussion. As shown in Fig. 12(a), the E-frac length decreased as the number of perforation clusters increased. This is because the greater number of fractures generated a superimposed stress shadow, thus rendering it more challenging to generate fractures as the number of clusters increases. Moreover, an increase in the number of perforation clusters under a fixed spacing between them resulted in a gradual increase in the distance between the first and final fractures. Consequently, a weaker stress shadow effect was achieved. The length of the multifracture decreased progressively as the number of perforation clusters increased. Under a fixed perforation spacing, an increase in the perforation clusters increased the distance between the first and final fractures, thus resulting in a weaker stress shadow effect. Similarly, the multifracture length decreased gradually as the number of perforation clusters increased. For the multifracture assemblies, the dimensions of the central fracture were smaller than those of the fractures on both sides. In the multifracture assemblies, the central fracture dimensions were consistently smaller than those on both sides. The variation in length within the multifracture assemblies was primarily due to the first and final fractures, which were more than double the size of the remaining fractures. This difference in the dimensions was caused by the spatial-location discrepancy of the hydraulic fractures. The central fractures exhibited the lowest total stress shadows.

The trend of the fracture-width variation with the number of perforation clusters presented in Fig. 12(b) is the same as that in shown Fig. 12(a), thus indicating that the fracture length is proportional to the width, and that their relationship is determined by parameters C and I in this model. In fact, the variation in the fracture length can be regarded as a variation in the fracture width. Therefore, the variation pattern of the fracture width will not be analysed in the following sections.

(2) Perforation cluster spacing

The fracture lengths under different perforation cluster spacings are shown in Fig. 13. When the number of perforation clusters was constant, an increase in the perforation cluster spacing reduced the total stress shadow, as expressed in Eqs. (32) and (33), thus increasing the net pressure and facilitating the propagation of hydraulic fractures. Therefore, the E-frac length increased with the perforation cluster spacing, as shown in Fig. 13(a). Fractures 2 and 3 exhibited the same variation trend as E-frac owing to the same reason. Notably, fractures 1 and 4 exhibited different variation trends. The fracture length initially decreased and subsequently increased as the perforation spacing increased. This is different from the results of Hu [22], which were not limited by the total fracture volume and the effect of natural fractures. Their results showed that the lengths of fractures 1 and 4 increased continuously as the stress shadowing decreased. Here, the lengths of Fractures 1 and 4 were significantly larger than those of fractures 2 and 3 when the perforation spacing was relatively small (5–15 m). Based on Eq. (26), a larger fracture length indicates a greater natural-fracture pressure drop. The restriction of natural fractures evidently supersedes the promotion of hydraulic fracture propagation via the perforation spacing. Consequently, the lengths of the first and final fractures exhibited different variation patterns at small perforation spacings. As the perforation cluster spacing increased, the dimensions of the multifracture became uniform.

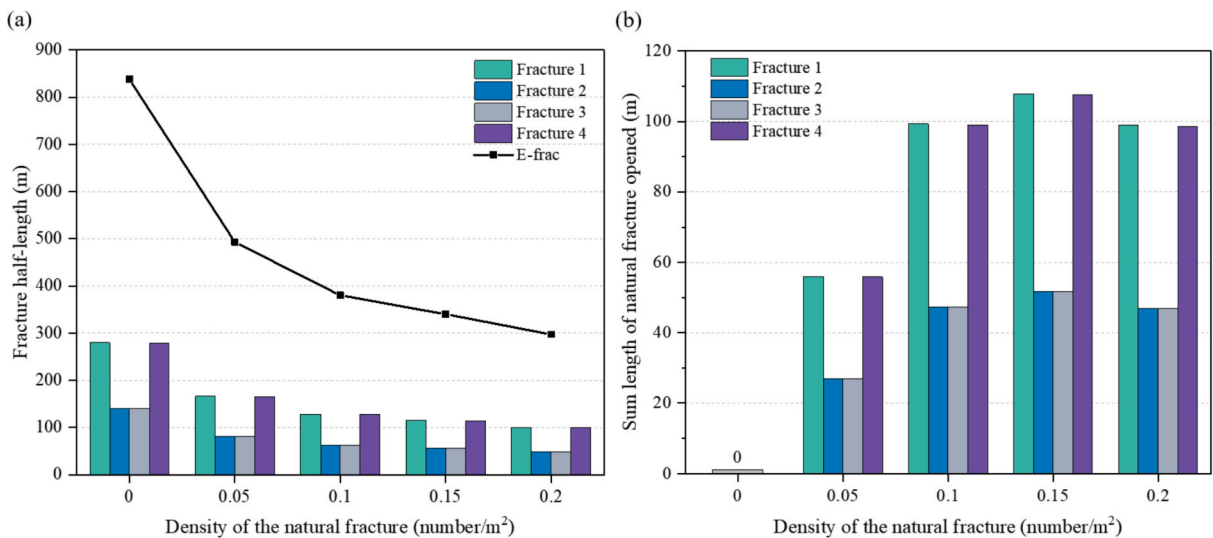


Fig. 14. Effect of density of natural fracture on hydraulic fracture dimension and natural-fracture opening: (a) Variation in hydraulic fracture, (b) Variation in opened natural fracture.

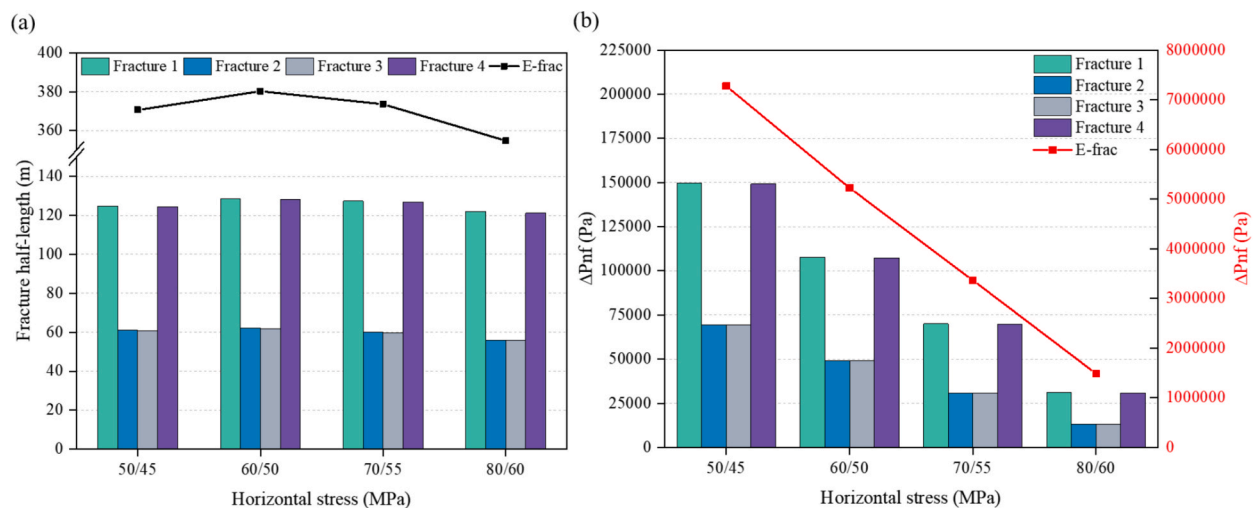


Fig. 15. Effect of density of horizontal stress on fracture dimension and natural-fracture pressure drop: (a) Variation of hydraulic fracture, (b) Variation in natural-fracture pressure drop.

Table 4

Input parameters for field case study.

Parameter	Value
Fracturing string size	5–1/2 in.
Depth of target reservoir	6102–6165 m
Steady pump rate	18 m ³ /min
Fracturing fluid viscosity	5.0 mPa · s
Fracturing fluid density	1050 kg/m ³
Speed of water hammer wave	1535.9 m/s
Carter leak-off coefficient	3.55×10^{-5} m/min ^{0.5}
Cluster number	6
Cluster spacing	10 m
Young's modulus of rock	37.4 GPa
Poisson's ratio of rock	0.22
Minimum horizontal stress	98 MPa
Maximum horizontal stress	109 MPa
Average length of natural fractures	7 m
Average width of natural fractures	5×10^{-5} m
Average strike angle of natural fractures	65°
Density of the natural fractures	0.012 number/m ²

3.2.2. Density of natural fracture

Fig. 14 illustrates the fracture dimensions with different densities of natural fracture (0–0.2 number/m²). As shown in Fig. 14(a), both the E-frac and multifracture decreased continuously as the density of natural fractures increased. This reduction is attributed to two primary factors: (1) enhanced leak-off of the fracturing fluid into the natural fractures, and (2) additional pressure drop caused by the increased number of natural fractures. The volume and pressure of the fracturing fluid that caused hydraulic fractures decreased. Consequently, the fracture dimensions decreased. Additionally, the fracture dimension in the naturally fractured reservoir was smaller than that in conventional reservoir. The variation gradients of fractures 1 and 4 were greater than those of fractures 2 and 3 because their larger fracture volumes strengthened the effect of leak-off and natural fractures. Meanwhile, whereas the natural fractures decreased the hydraulic fracture length, more natural fractures opened and propagated as their density increased.

It is shown that while increasing natural fracture density reduces the length of hydraulic fractures, it also promotes the opening and propagation of more natural fractures. This resulted in a hydraulic fracture with a complex fracture network, which implies that a more stimulated volume is available for fracturing treatment. However, as the hydraulic fracture length decreases, the probability of opening a natural fracture decreases as well. The sum of the opened natural fractures decreased at a density of 0.2 number/m², as shown in Fig. 14(b). The other observation is that the naturally fractured reservoir resulted in more complex hydraulic fractures with an additional volume of opened natural fractures, and that the development of natural fractures in the reservoir did not increase the complexity of hydraulic fractures indefinitely. It is because that higher fracture resistance plays a critical role in fracture propagation dynamics. Specifically, higher resistance in natural fractures means that more energy is required to open and propagate them. Consequently, more natural fractures are opened, but the hydraulic fracture dimensions become smaller due to the increased energy dissipation.

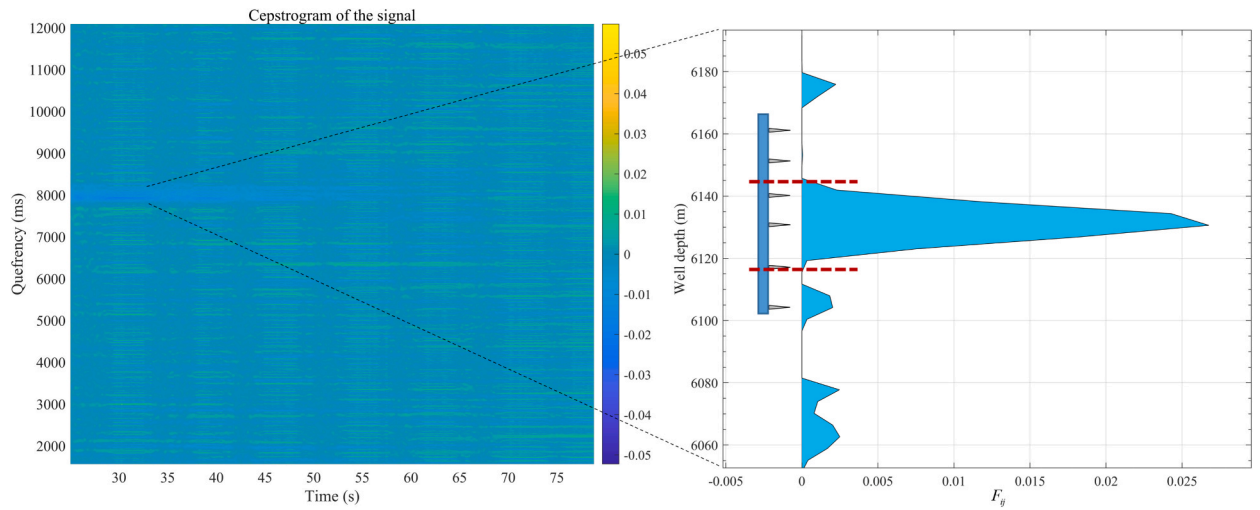


Fig. 16. Water-hammer signal cepstrum for field case study.

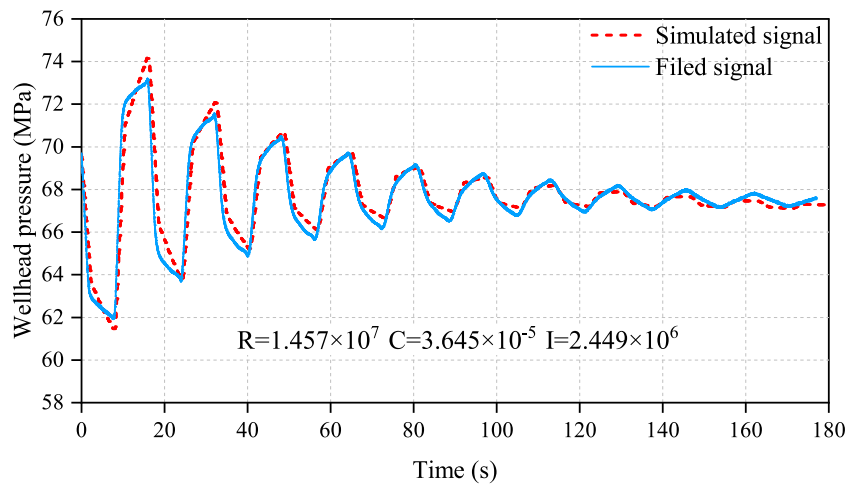


Fig. 17. Simulated and field water-hammer pressure matching results.

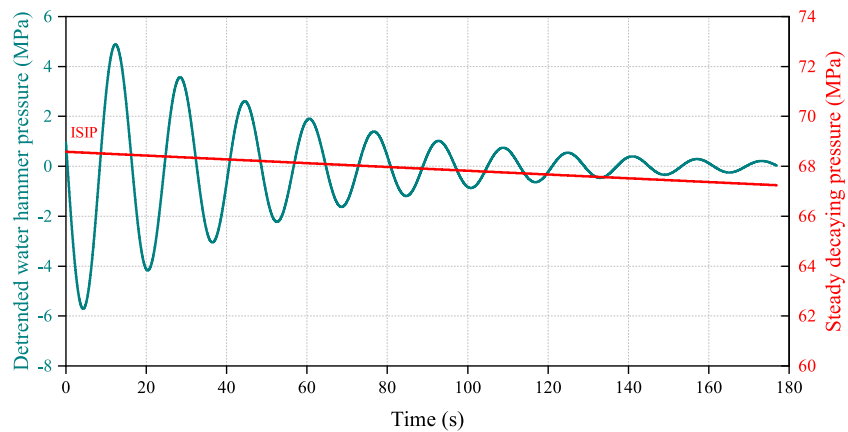


Fig. 18. Water-hammer pressure-decomposition result.

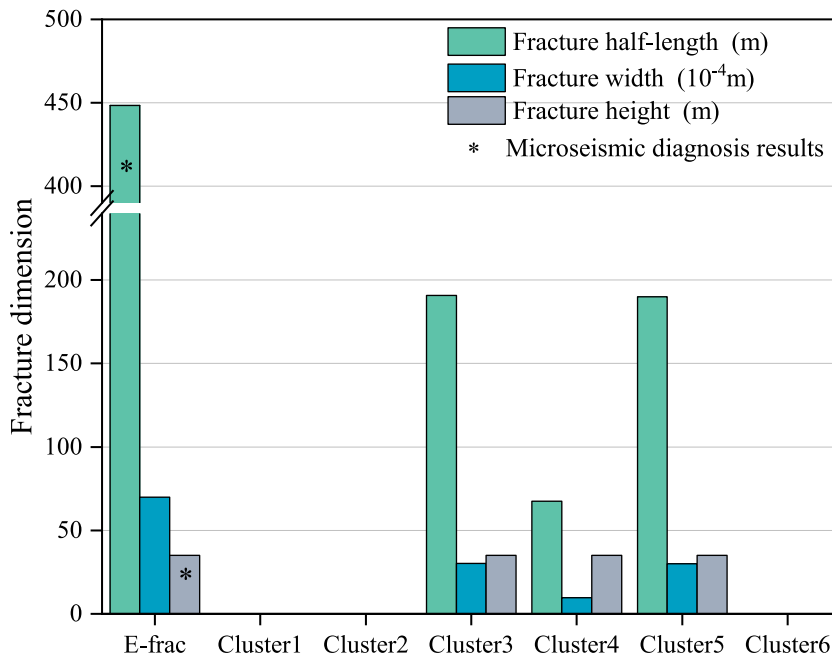


Fig. 19. Equivalent fracture and multifracture dimension diagnostic results.

3.2.3. Horizontal stress

Horizontal stress is the main factor that must be overcome to ensure the propagation of hydraulic fractures. The minimum horizontal stress determines the net fracture pressure, and the horizontal stress difference affects the interaction mode between the hydraulic and natural fractures. In this section, we investigate the fracture dimensions under different horizontal stresses, with the maximum/minimum horizontal stresses set to 50/45, 60/50, 70/55, and 80/60 MPa. The minimum horizontal stress and horizontal stress difference were increased progressively for the calculation cases. The results are shown in Fig. 15. The fracture length initially increased and then decreased. This phenomenon is attributable to the alteration in the net fracture pressure caused by the conjunction of the minimum horizontal stress and natural fracture pressure drop, which can be determined using Eqs. (31) and (32), respectively. An increase in the minimum horizontal stress resulted in a shorter fracture length. Meanwhile, a greater horizontal stress difference facilitated the propagation of hydraulic fractures across natural fractures, thus resulting in fewer opened and propagated natural fractures and less additional pressure drops from natural fractures, as depicted in Fig. 15(b). This phenomenon facilitates the realisation of larger fracture dimensions. In addition, the effect of the minimum horizontal stress on the fracture length was more severe than that of the horizontal stress difference under a high minimum horizontal stress. For instance, the average natural fracture pressure drop of the E-frac was only 1.49 MPa, with a minimum horizontal stress and horizontal stress difference of 60 and 20 MPa, respectively. Therefore, reservoir stress must be considered in the diagnosis of fractured reservoir.

3.3. Field case study

The multifracture diagnostic model was evaluated for its application and performance during the completion of shale horizontal-well fracturing in the Sichuan Basin, China. A staged multicluster fracturing technique was employed in the horizontal section. Logging and microseismic monitoring data indicated the development of natural fractures in the reservoir. Meanwhile, a staged multicluster fracturing technique was employed in the horizontal section, in which the interval was uniformly perforated with a 10 m cluster spacing and then isolated with mechanical bridge plugs. A high frequency pressure sensor was installed at the wellhead to obtain the water-hammer signal during fracturing shut-in. The main input parameters of the diagnostic model are listed in Table 4. The theoretical calculation of the water-hammer wave speed was performed according to Eq. (7). The water-hammer wave speed was also adjusted by the period time of occurrence of the water-hammer pulsation at the wellhead.

Initially, the number and location of hydraulic fractures were analysed via CEEM enhancement and a cepstrum process on a high frequency water-hammer signal. The well was designed with six perforation clusters at a depth of 6102–6165 m. The stimulation result showed negative peaks appearing at the frequency area of 7964–8004 ms (depth of 6116–6147 m), as shown in Fig. 16. This suggests that the perforation clusters placed within the depths of 6116–6147 m absorbed the fracturing fluid, whereas other clusters remained unopened, thereby inhibiting the creation of hydraulic fractures. Thus, the three perforation clusters between 6116–6147 m were regarded as the quantity and placement of hydraulic fractures. Later, these data will be used to assess the stress shadow and fracture dimensions.

The water-hammer signal from the field (blue curve in Fig. 17) was decomposed into detrended and steady decaying pressures to

obtain the *ISIP* and boundary conditions for the water-hammer pressure simulation, as shown in Fig. 18. The detrended water-hammer pressure were approximately zero, thus indicating energy oscillation and a decay in the wellbore. Notably, a steadily decreasing pressure (red curve in Fig. 17) was achieved by removing the effect caused by the friction of the water-hammer flow, which characterises the variation in the actual hydraulic fracture pressure over time. Thus, a steadily decreasing pressure was employed as a boundary condition in the subsequent simulation, and the *ISIP* was obtained from the red-curve intercept. In addition, the water-hammer signals were history matched and compared. The values of *R*, *C*, and *I* were iteratively changed until the simulated data were similar to the field water-hammer signal obtained at the wellhead. The matching results are presented in Fig. 17. The maximum error was less than 1 MPa. The match quality in terms of period, amplitude, and decay rate was acceptable. The components at each end of the signal showed less satisfactory matching owing to the quality of the water-hammer signal and assumptions introduced regarding the flow rate changes at the wellhead. The *R*, *C*, and *I* were identified as $1.457 \times 10^7 \text{ Pa} \cdot \text{s}/\text{m}^3$, $3.645 \times 10^{-5} \text{ m}^3/\text{Pa}$, and $2.449 \times 10^6 \text{ Pa}/\text{m}^3 \cdot \text{s}^{-2}$, respectively.

After determining the fracture location, number, and *RCI*, the E-frac and multifracture dimensions were estimated, and the results are presented in Fig. 19. The half-length, width, and height of the E-frac were 448.3, 70×10^{-6} , and 35 m, respectively. The relative errors for the half-length and height were 6.5 % and 20 %, respectively, compared with the diagnostic results of the fracture length and height obtained from microseismic analysis—the magnitudes of these error are regarded as acceptable for field-construction analysis. The limited diagnostic accuracy of the fracture height is attributed to its confinement within the boundaries of the reservoir thickness, which will be addressed in the subsequent improvement of our model. Based on the analysis presented in Section 3.2, the fracture dimension on both sides tend to propagate more easily than that at the centre because of the stress shadow. The half-lengths of the three propagated fractures were 190.8, 67.5, and 190.0 m. The maximum difference in the multifracture half-length was 123.3 m, which was greater than the average fracture half-length. This shows that previous fracturing optimisation design studies were ineffective in promoting the uniform propagation of fractures. Meanwhile, the diagnostic results showed that clusters 1, 2, and 6 were not opened to propagate hydraulic fractures, thus indicating that the reservoir was not stimulated appropriately. Therefore, measures such as injecting a temporary plugging material should be implemented to improve the stimulation effect of the well. Additionally, the diagnostic results provide guidance for subsequent fracturing optimisation design.

4. Conclusion

In this study, a multifracture *RCI* diagnostic model for naturally fractured reservoir was proposed. To address some of the limitations of the previous model, we employed the cepstrum of water hammer in our model to obtain the location and number of effectively propagated hydraulic fractures as well as considers the effects of leak-off and natural fractures. Wellhead water-hammer pressures with different fracturing shut-in modes and multifracture dimensions with different perforation designs, densities of natural fractures, and horizontal stresses were investigated. Finally, a case study on a field shale well was performed to demonstrate the application process and performance of our model. The primary conclusions obtained were as follows:

- (1) The water-hammer pressure pulsation was related to the shut-in time and displacement. A staged pump shutdown with extended shut-in times effectively prevented large pressure pulsations that damage the wellbore integrity. Similarly, the fracturing shut-in method can be appropriately modified to obtain a clear water-hammer signal for fracturing diagnostics.
- (2) As the number of shot clusters increased, the fracture length and width decreased gradually, and the dimensions of the fracture in the centre were consistently smaller than those on both sides. As the perforation cluster spacing increased, the fracture on both sides decreased and then increased, the centre fracture increased continuously, and the dimensions of the multifracture eventually became uniform.
- (3) Natural fractures can reduce hydraulic fracture dimensions. However, complex fracture networks should be created to increase the stimulated volume. The development of natural fractures in reservoir does not indefinitely increase the complexity of hydraulic fractures because the probability of natural fracture propagation is limited by the hydraulic fracture dimensions.
- (4) An increase in the minimum horizontal stress decreased the fracture dimensions. Meanwhile, a greater difference in the horizontal stress difference rendered it easier for a hydraulic fracture across a natural fracture to create a larger fracture dimension.
- (5) Field fracture diagnostic results showed three propagated fractures at the locations of perforation clusters 3, 4, and 5. The fracture diagnostic dimensions are acceptable, based on a comparison with microseismic results, and thus can be used to improve the fracturing design.

CRedit authorship contribution statement

Shijie Deng: Writing – original draft, Methodology, Investigation. **Liangping Yi:** Writing – review & editing, Funding acquisition, Formal analysis. **Xiaogang Li:** Methodology, Investigation. **Zhaozhong Yang:** Resources, Conceptualization. **Nanqiao Zhang:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was financially supported by the National Natural Science Foundation of China (42202314).

Appendix A

The steps involved in the CEEMD process, which is shown in Fig. A 1, are as follows:

- (1) Input the water-hammer pressure signal and add white noise.
- (2) Determine the points of the maximum and minimum signal values, plot the maximum–minimum envelope, and calculate the mean envelope $m(t)$.
- (3) Obtain the suspected IMF component $h(t)$ by subtracting the average envelope $m(t)$ from the original signal $x(t)$.
- (4) Determine whether $h(t)$ satisfies the IMF component conditions. If the average value of m^t tends to zero and the difference between the number of extreme points of the original signal and the number of intersection points of the original signal with the x-axis is not greater than 1, then extract the IMF and residue $r(t)$, otherwise, $x(t) = h(t)$, and return to Step (2).
- (5) Test whether the residue satisfies the stopping condition. If the residue satisfies the stopping condition, then output the IMF component, otherwise $s(t) = r(t)$ and returns to Step (1).

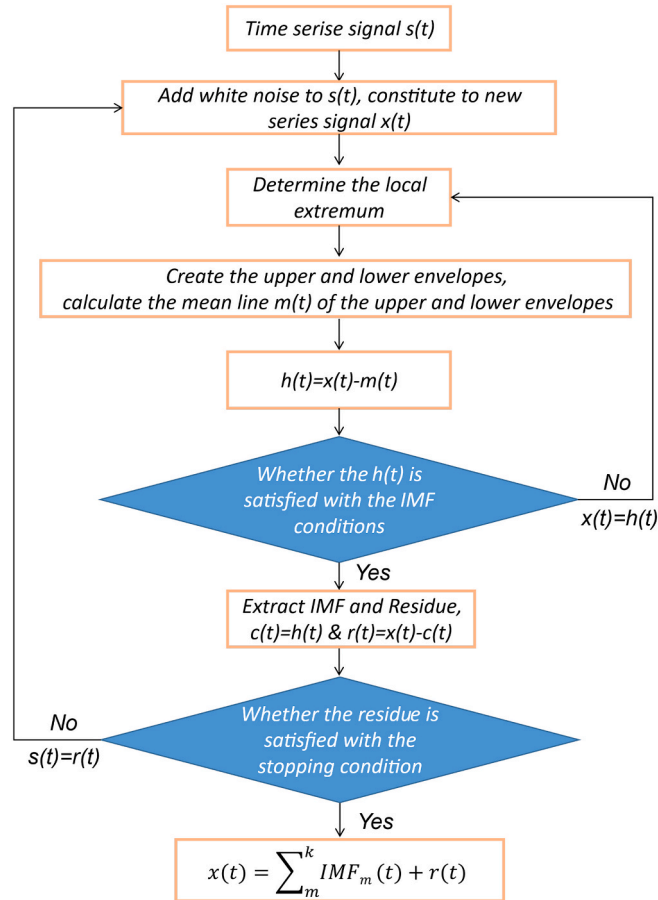


Fig. A1. CEEMD process of water-hammer pressure signal.

Appendix B

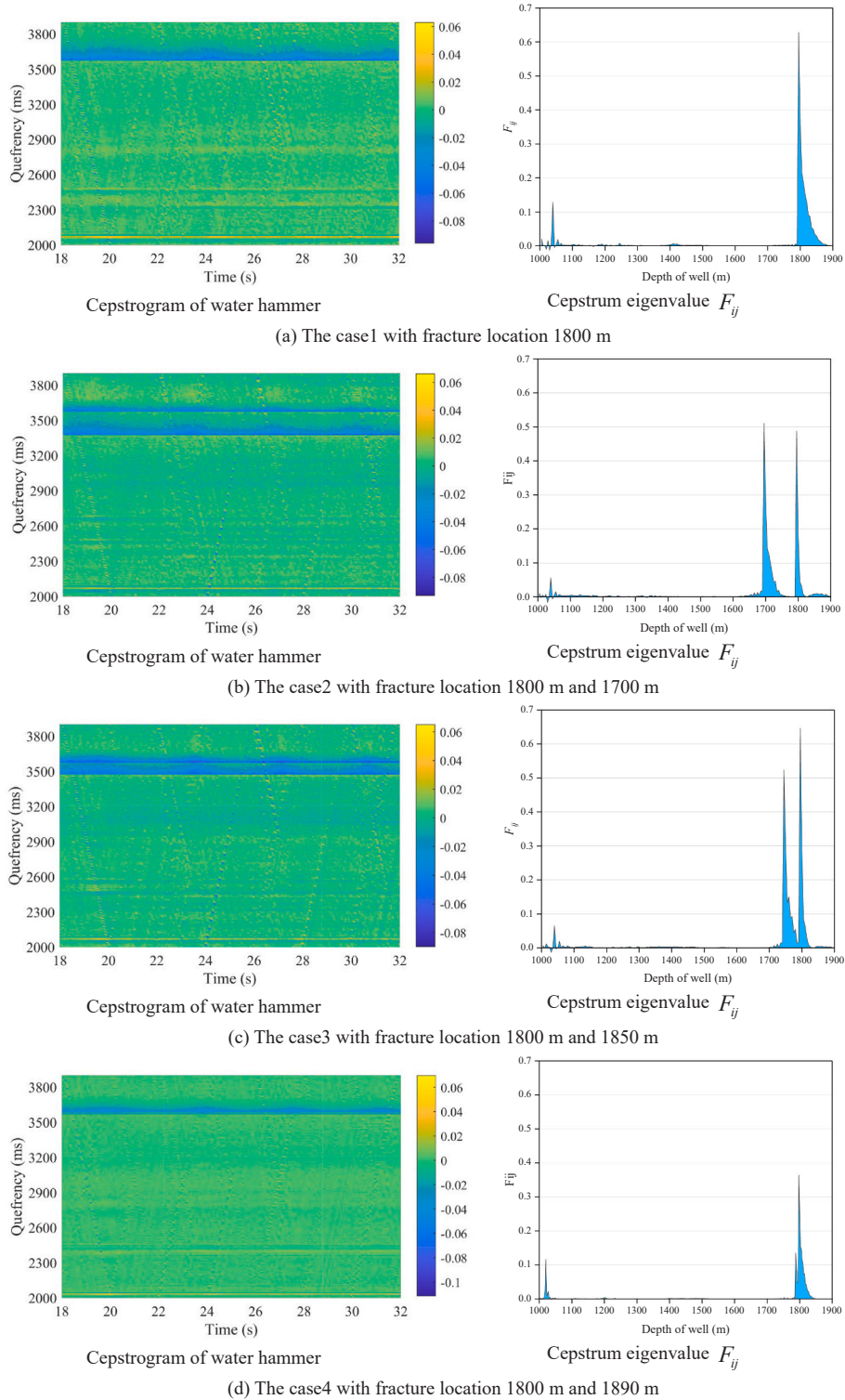


Fig. B1. Diagnostic results of with different artificially fracture setting.

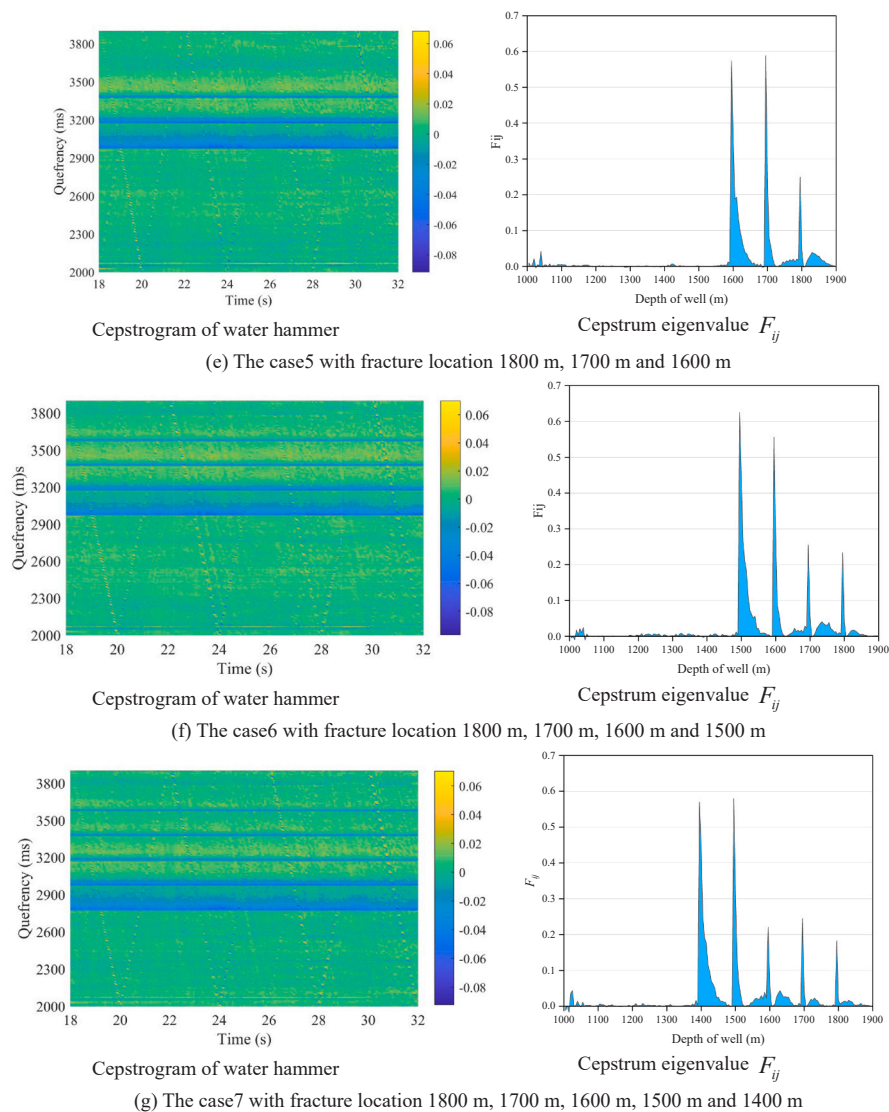


Fig. B1. (continued).

Data availability

Data will be made available on request.

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